COURSE NOTES

Combinatorics II

Instructor:

Prof. Ferdinand IHRINGER

Student:

Leo

2026 年 3 月 26 日

目录

第一章 Lecture 1

Basic Information

Office hours: Mon, 13:45-15:45.

Main book: *A course in combinatorics* by J.H. van Lint and R.M. Wilson, 2nd edition. And the lecture notes by Prof. Ihringer.

Grade: Final Exam: 40%, midterm 30 % (April 15, week 8 in class), Homework 20%, Quizzes 10% (very easy based on HW).

Homework:

- Biweekly
- Only handwritten on paper
- Submission in class
- Only in English

Preliminary Schedule:

- **Week 1-4:** Basic graph theory, including trees, Ramsey numbers, Turan numbers, Hall's marriage theorem, etc.
- **Week 5-6:** Hadamard matrices.
- **Week 6-8:** Projective spaces.
- **Week 9:** Designs.
- **Week 10:** Strongly regular graphs.
- **Week 11:** Generalized Quadrangles.
- **Week 12-13:** General Spectral Graph Theory.
- **Week 14:** Associated Schemes.
- **Week 15:** Coding Theory.

1.1 Introduction to Graphs

Definition 1.1.1. *A graph G is a pair (V, E) where V is a set of vertices and E is a set of edges.*

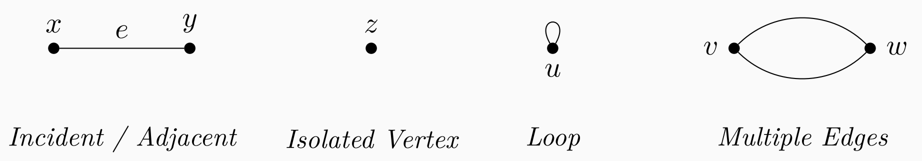
Let $G = (V, E)$ be a graph with vertex set V and edge set E.

- If G is directed, then edges are ordered pairs and $E \subseteq V \times V$;
- If G is simple undirected, then edges are unordered pairs and $E \subseteq \binom{V}{2}$. (We call such G a normal graph in our course.)

Definition 1.1.2. *Let $G = (V, E)$ be a graph.*

- *Let $e = \{x, y\} \in E$. The vertices x and y are said to be **incident** with the edge e. Two vertices x, y are called **adjacent** if $\{x, y\} \in E$.*
- *A vertex is called an **isolated vertex** if it is not incident with any edge.*
- *An edge connecting a vertex to itself is called a **loop**.*
- *Two or more edges connecting the same pair of vertices are called **multiple edges**.*

Example 1.1.3. *The following figure illustrates the definitions:*



- ***Incident/Adjacent**: Vertices x and y are adjacent and incident to edge e.*
- ***Isolated**: Vertex z has no incident edges.*
- ***Loop**: Vertex u has an edge connecting to itself.*
- ***Multiple Edges**: Vertices v and w are connected by two edges.*

Definition 1.1.4. *A graph is called a **planar graph** if it possesses a planar drawing, that is, it can be drawn in the plane such that no two edges intersect except at their endpoints.*

Example 1.1.5. *Consider the following two drawings:*

*G_1 and G_2 are **isomorphic**: If we pull the center node of G_1 then we get G_2. G_1 is planar, but G_2 is not. We call G_1 the plannar drawing of G_2, which means that G_1 has no edges intersect.*

Definition 1.1.6. *A graph without loops and multiple edges is called a **simple graph**.*

Remark 1.1.7. *In the two statements above, the intended notion is essentially the same. The condition "G is simple undirected" together with $E \subseteq \binom{V}{2}$ means that every edge is an unordered pair $\{u, v\}$ of* distinct *vertices, so loops are excluded automatically, and since E is a set (rather than a multiset), multiple edges are excluded as well. Hence this coincides with the standard definition:* a simple graph *is a graph with no loops and no multiple edges.*

In our course notes, such graphs are called normal graphs. *The only minor caveat is that some texts use "simple graph" with the convention that graphs are undirected by default, whereas others may state "simple undirected" explicitly. In the present context, these conventions agree.*

Convention: All our graphs are simple. Unless we say otherwise.

Definition 1.1.8 (Map coloring and the associated graph)**.** *A* map *(on the plane or on the sphere) is a decomposition of the surface into finitely many regions ("countries") such that any two regions meet either in a boundary arc, in finitely many boundary points, or not at all. Two countries are said to be* adjacent *if they share a boundary segment of positive length (meeting at a single point does* not *count).*
Given such a map, we define its adjacency graph *(also called the* dual graph*) $G = (V, E)$ as follows:*

$$V = \{\textit{countries of the map}\}, \qquad \{u, v\} \in E \iff \textit{countries } u \textit{ and } v \textit{ are adjacent.}$$

Remark 1.1.9 (Planarity)**.** *The adjacency graph G is* planar*: it can be drawn in the plane with no edge crossings by placing a vertex inside each country and drawing an edge between two vertices through the common boundary segment of the corresponding countries. Thus the problem of coloring maps is equivalent to coloring planar graphs.*

Definition 1.1.10 (Vertex coloring and chromatic number)**.** *Let $G = (V, E)$ be a graph. A* (proper) vertex coloring *of G with k colors is a map $c : V \to \{1, 2, \dots, k\}$ such that*

$$\{u, v\} \in E \implies c(u) \neq c(v).$$

The smallest k for which such a coloring exists is called the chromatic number *of G and is denoted by $\chi(G)$.*

Theorem 1.1.11 (Four-Color Theorem)**.** *For every planar graph G, one has*

$$\chi(G) \leq 4.$$

Equivalently, every planar map can be colored with at most four colors so that any two adjacent countries receive different colors.

Remark 1.1.12 (Why "at most 5 colors" is easy)**.** *A classical result (the* Five-Color Theorem*) states that every planar graph satisfies $\chi(G) \leq 5$. This theorem admits a relatively short proof using standard planar graph tools such as Euler's formula and a reducible configuration argument (often presented via induction on $|V|$). By contrast, the Four-Color Theorem is substantially deeper and historically required extensive case-checking (computer-assisted) in its first proofs.*

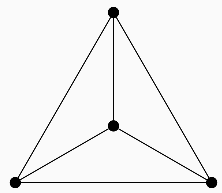
The following statements are equivalent:

$$\{x, y\} \in E \Leftrightarrow x \sim y \Leftrightarrow x, y \text{ adjacent } \Leftrightarrow x, y \text{ are neighbors}$$

Example 1.1.13. $|V| = n$, $E = \binom{V}{2}$. *Complete graph K_n.*

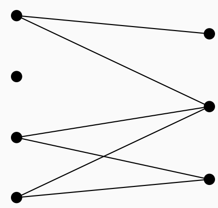
K_1 K_2 K_3 K_4 K_5

Remark: K_4 is a planar graph, but K_5 is not.

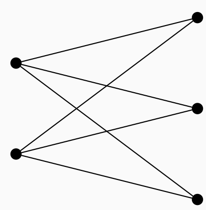


Planar drawing of K_4

Example 1.1.14. $G = (V_1 \cup V_2, E)$ *is bipartite if* $E \cap (\binom{V_1}{2} \cup \binom{V_2}{2}) = \emptyset$.

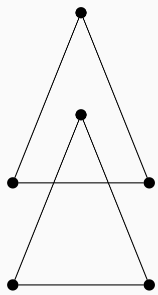


Example 1.1.15. *If $E = \{\{x, y\} : x \in V_1, y \in V_2\}$, then G is a complete bipartite graph, denoted by $K_{m,n}$ where $m = |V_1|, n = |V_2|$.*

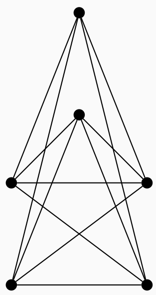


$K_{2,3}$

Example 1.1.16. *Similarly we can define k-partite graphs and complete k-partite graphs.*



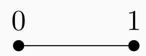
3-partite (2,2,2)
(not complete)



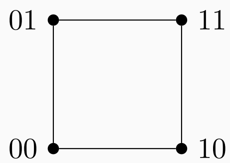
$K_{2,2,2}$

Example 1.1.17. *The n-**dimensional hypercube** Q_n is the graph with vertex set $V(Q_n) = \{0,1\}^n$, the set of binary strings of length n. Two vertices $x, y \in V(Q_n)$ are adjacent if and only if they differ in exactly one coordinate (i.e., their Hamming distance is 1). The graph Q_n is n-regular with $|V(Q_n)| = 2^n$ and $|E(Q_n)| = n2^{n-1}$. For $n = 3$, we have $110 \sim 111$, whereas $111 \nsim 100$.*

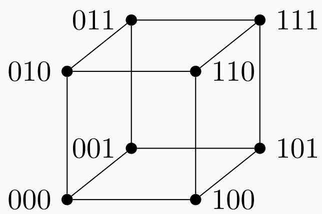
Q_0



Q_1

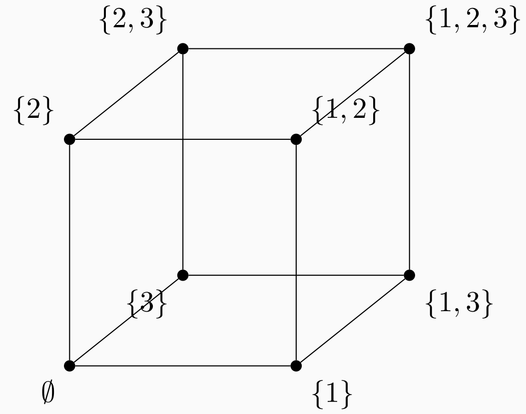


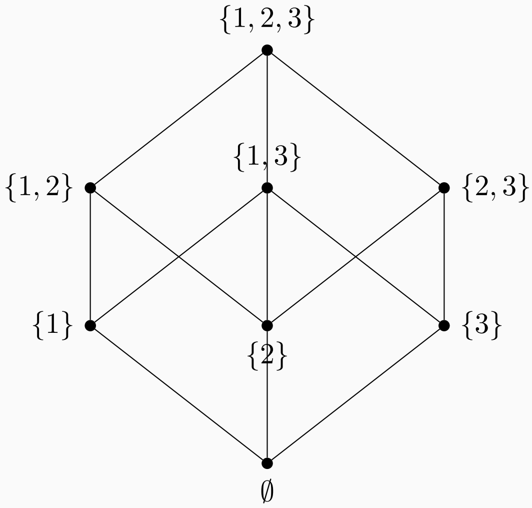
Q_2



Q_3

Example 1.1.18. *Let S be a set with 3 elements, for instance $S = \{1, 2, 3\}$. The power set $\mathcal{P}(S)$ can be made into a graph by defining two subsets $A, B \subseteq S$ to be adjacent if their symmetric difference has size one, i.e., $|A \Delta B| = 1$. This graph is isomorphic to the 3-dimensional hypercube Q_3. The structure is often drawn as the Hasse diagram of the subset lattice.*

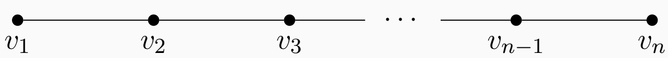




Hasse lattice of $\mathcal{P}(\{1,2,3\})$

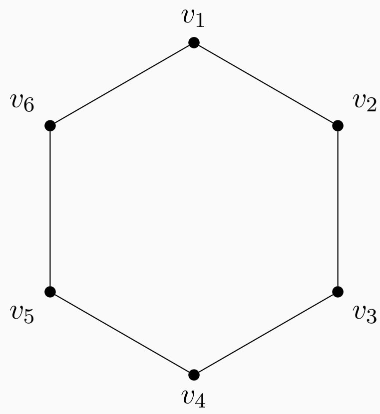
This graph is another representation of Q_3.

Example 1.1.19. P_n*: path of length* $n-1$.



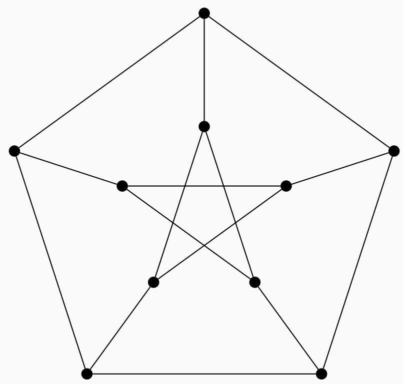
$$|E(P_n)| = n-1$$

Example 1.1.20. *With* $\{v_n, v_1\}$ *added, we get a cycle* C_n. *For example* C_6:



$|V(C_6)| = |E(C_6)| = 6$

Example 1.1.21. *Peterson Graph:*



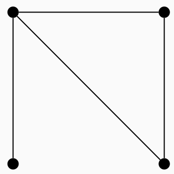
Petersen graph

Each vertex has degree 3 so this graph is 3*-regular. Any two vertices linked by an edge has no common neighbor. Any 2 vertices not linked by an edge has a common neighbor.*
It is a strongly regular graph with parameters $(10, 3, 0, 1)$.

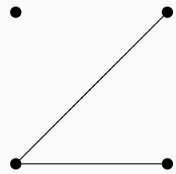
Definition 1.1.22 (Strongly regular graph)**.** *Let* $G = (V, E)$ *be a simple undirected graph. We say that* G *is* strongly regular *with parameters* (v, k, λ, μ) *if the following hold:*

1. $|V| = v$ *and* G *is* k*-regular (i.e. every vertex has degree* k*);*
2. *for any two* adjacent *vertices* $x \sim y$*, the number of common neighbors of* x *and* y *is exactly* λ*, i.e.* $|\Gamma(x) \cap \Gamma(y)| = \lambda$*;*
3. *for any two* non-adjacent *vertices* $x \not\sim y$*, the number of common neighbors of* x *and* y *is exactly* μ*, i.e.* $|\Gamma(x) \cap \Gamma(y)| = \mu$.

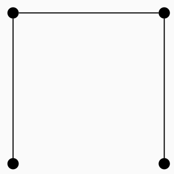
Example 1.1.23. *$G=(V,E)$, $\bar{G}=(V,\bar{E})$, where $\bar{E}=\binom{V}{2}\setminus E$. $\bar{G}$ is the complement of G.*



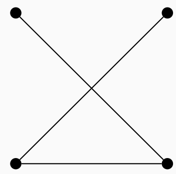
G



$\overline{G}$



P_4



$\overline{P_4}$

In this example, observe that $P_4 \cong \overline{P_4}$. We call such a graph self-complementary. A graph G is self-complementary if $G \cong \overline{G}$.

Definition 1.1.24. *Let $G=(V,E)$ and $G'=(V',E')$. They are isomorphic if and only if there exists a bijection $\phi: V \to V'$ such that $\{x,y\} \in E \Leftrightarrow \{\phi(x),\phi(y)\} \in E'$.*

Definition 1.1.25. *$\sharp y : y \sim x$ = degree of x, denoted by $\deg(x)$ or $\mathrm{d}(x)$. If $\mathrm{d}(x) = \mathrm{d}(y)$ for all $x,y \in V$, then G is d-regular.*

Remark 1.1.26 (Degree sequence: definition and uses)**.** *Let $G=(V,E)$ be a simple undirected graph with $n=|V|$. The* degree sequence *of G is the list of vertex degrees arranged in nonincreasing order,*

$$d_1 \geq d_2 \geq \cdots \geq d_n, \qquad \text{where } \{d_1,\ldots,d_n\} = \{\deg(v) : v \in V\}.$$

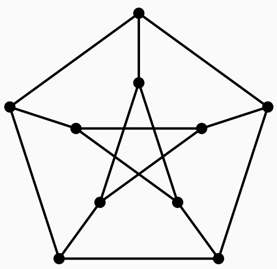
It is useful for:

- ***Coarse structural information:*** *by the Handshaking Lemma, $\sum_{i=1}^{n} d_i = 2|E|$, so the degree sequence determines the number of edges and reveals properties such as regularity (all d_i equal).*
- ***Realizability tests:*** *deciding whether a given integer sequence is* graphical *(i.e. occurs as the degree sequence of a simple graph), using criteria such as the Erdős–Gallai theorem or the Havel–Hakimi algorithm (which can also construct a realizing graph).*

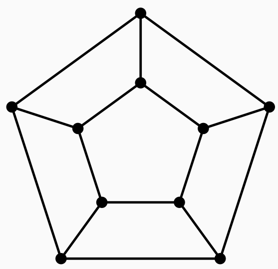
- ***Isomorphism invariant:*** *isomorphic graphs have the same degree sequence, so differing degree sequences imply non-isomorphism, though the converse need not hold.*
- ***Algorithmic and modeling input:*** *degree sequences serve as compact summaries in extremal graph theory, network models (e.g. random graphs with prescribed degrees), and matrix-based methods (e.g. Laplacian $L = D - A$ where D is the degree matrix).*

Example 1.1.27. $d_1 = 2, d_2 = 1, d_3 = 1$. *This graph is* P_3.

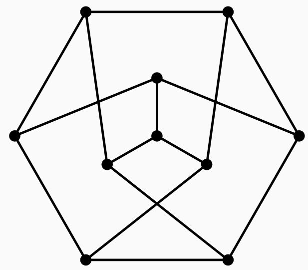
Example 1.1.28. *Consider the following 3 graphs:*



G_1



G_2



G_3

$G_1 \ncong G_2$, $G_2 \ncong G_3$, *but* $G_1 \cong G_3$.

The *graph isomorphism problem* (GI) asks: given two finite graphs G and H, decide whether there exists a bijection between their vertex sets that preserves adjacency (i.e. an isomorphism). From the viewpoint of worst-case complexity, GI has long occupied an intermediate position: it is in NP, but it is neither known to be NP-complete nor known to admit a polynomial-time algorithm. A major milestone is Babai's 2015 breakthrough, which gives a *quasi-polynomial* time algorithm for GI, running in time $\exp\big((\log n)^{O(1)}\big)$ on n-vertex graphs, substantially improving the best previously known general upper bounds.

In parallel with the worst-case theory, there is a large body of *practical* work aimed at solving typical instances efficiently, often via *canonical labeling* and computation of automorphism groups. Since the 1990s, McKay's software `nauty` (and later `Traces`, with further algorithmic refinements) has been highly effective on a wide range of graphs encountered in applications. However, strong empirical performance does not settle the worst-case complexity question, and there exist carefully constructed families of graphs that can be difficult for certain refinement-and-search heuristics. Thus, it is important to distinguish clearly between (i) the unresolved *theoretical* question of whether GI is solvable in polynomial time in the worst case, and (ii) the largely successful but still evolving *algorithm engineering* of fast GI solvers for real-world inputs.

Proposition 1.1.29 (handshaking lemma). $G = (V, E)$. *Then*

$$\sum_{v \in V} \deg(v) = 2|E|$$

证明. Count $(x, e) \in V \times E$ where $x \in e$ in two ways. First, for each $e \in E$, there are exactly two vertices $x \in e$. So the number of such pairs is $2|E|$. Second, for each $x \in V$, there are exactly $\deg(x)$ edges $e \in E$ containing x. So the number of such pairs is $\sum_{v \in V} \deg(v)$. Equating these two counts gives the desired result. □

Theorem 1.1.30. *A finite graph (we only talk about finite graphs in our class) has an even number of vertices with odd degree.*

证明. Consider

$$\sum_{v \in V} \deg(v) = 2|E|,$$

which is an even integer.

Reduce this congruence modulo 2. Vertices of even degree contribute 0 modulo 2, and vertices of odd degree contribute 1 modulo 2. Hence

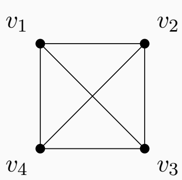
$$\sum_{v \in V} \deg(v) \equiv \#\{v \in V : \deg(v) \text{ is odd}\} \pmod 2.$$

Since the left-hand side is even, the right-hand side is $0 \pmod 2$, so the number of odd-degree vertices is even. □

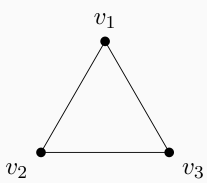
Definition 1.1.31 (subgraph). *Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is a* subgraph *of G if $V' \subseteq V$ and $E' \subseteq E$.*

Definition 1.1.32 (induced subgraph). *Let $G = (V, E)$ be a graph. A graph $H = (V', E')$ is an* induced subgraph *of G if $V' \subseteq V$ and $E' = E \cap \binom{V'}{2}$.*

Example 1.1.33. *Consider the graph K_4, we have two subgraph C_3 and C_4. The subgraph C_3 is an induced subgraph, but the subgraph C_4 is not.*

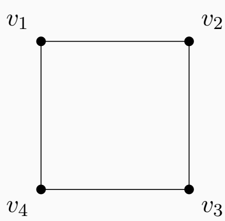


K_4



C_3

induced subgraph



C_4

not induced in K_4

For C_3, we choose the vertices $\{v_1, v_2, v_3\}$, and every edge among them in K_4 is kept, *so it is induced. For C_4, we use all four vertices but delete the two diagonals, so it is a subgraph but not an induced subgraph.*

Adjacency list representation.

Let $G = (V, E)$ be a (finite) graph with $V = \{1, 2, \ldots, n\}$. In an *adjacency list* representation, the computer stores an array (or map) Adj indexed by vertices, where $\mathrm{Adj}[u]$ is the list of all neighbors of u:

$$\mathrm{Adj}[u] = \{\, v \in V : \{u, v\} \in E \,\} \quad \text{(undirected)}, \mathrm{Adj}[u] = \{\, v \in V : (u, v) \in E \,\} \quad \text{(directed)}.$$

For an undirected simple graph each edge $\{u, v\}$ appears twice, once in $\mathrm{Adj}[u]$ and once in $\mathrm{Adj}[v]$. This data structure supports efficient traversal of neighbors (e.g. BFS/DFS) in total time $O(|V| + |E|)$ and uses space $O(|V| + |E|)$ for sparse graphs.

For our human beings we use the following matrices:

Definition 1.1.34 (adjacent matrix)**.** *Let $G = (V, E)$ be a graph. Let $V = \{v_1, v_2, \ldots, v_n\}$ and $E = \{e_1, e_2, \ldots, e_m\}$. Let $A = (a_{ij})$ where $a_{ij} = 1$ if $v_i \sim v_j$, and $a_{ij} = 0$ otherwise. Then A is called the* adjacent matrix *of G. For the base field usually we use $\mathbb{R}$, $\mathbb{C}$, $\mathbb{F}_2$, $\mathbb{F}_q$.*

Remark 1.1.35. *In our class we mainly use consider $A \in \mathbb{R}^{n\times n}$. Sometimes we will consider $A \in \mathbb{C}^{n\times n}$ when we move to spectral graph theory.*

Definition 1.1.36 (Incidence matrix)**.** *Let $B = (b_{ij})$, $B \in \mathbb{R}^{n\times m}$. $b_{ij} = 1$ if vertex v_i is incident to edge e_j, and $b_{ij} = 0$ otherwise. Then B is called the* incidence matrix *of G.*

Definition 1.1.37 (degree matrix)**.** *Let $D = (d_{ij})$ be a diagonal matrix where $d_{ii} = \deg(v_i)$, and $d_{ij} = 0$ if $i \neq j$. Then D is called the* degree matrix *of G.*

Exercise 1.1.38. *Prove that $A = BB^T - D$.*

证明. We assume throughout that $G = (V, E)$ is a finite *simple* undirected graph (no loops, no multiple edges) and that B is the (non-oriented) incidence matrix defined by

$$b_{ij} = \begin{cases} 1, & \text{if } v_i \text{ is incident to } e_j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $M := BB^T = (m_{ik})$. Then

$$m_{ik} = (BB^T)_{ik} = \sum_{j=1}^{m} b_{ij}\, b_{kj}.$$

For fixed vertices v_i, v_k, the product $b_{ij}b_{kj}$ equals 1 precisely when *both* v_i and v_k are incident to the same edge e_j, and equals 0 otherwise. Hence m_{ik} counts the number of edges that are incident to both v_i and v_k.

Case 1: $i = k$**.** Then m_{ii} counts the number of edges incident to v_i, so

$$m_{ii} = \deg(v_i) = d_{ii}.$$

Case 2: $i \neq k$**.** In a simple graph, there is *at most one* edge joining v_i and v_k. Therefore

$$m_{ik} = \begin{cases} 1, & \text{if } v_i \sim v_k, \\ 0, & \text{otherwise,} \end{cases}$$

so for $i \neq k$ we have $m_{ik} = a_{ik}$.

Combining the two cases, we see that BB^T agrees with A on all off-diagonal entries, and has diagonal entries $\deg(v_i)$. Equivalently,

$$BB^T = A + D,$$

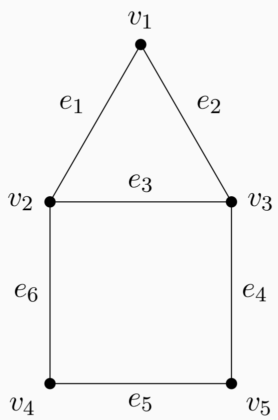
and therefore

$$A = BB^T - D,$$

as required. □

Definition 1.1.39 (Laplacian matrix)**.** *Let $L = D - A$, where D is the degree matrix and A is the adjacency matrix. Then L is called the* Laplacian matrix *of G.*

Example 1.1.40. *Consider this graph:*



With the vertex order $(v_1, v_2, v_3, v_4, v_5)$ and the edge order $(e_1, e_2, e_3, e_4, e_5, e_6)$, the *corresponding matrices are:*

$$A = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

$$L = D - A = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 \\ -1 & -1 & 3 & 0 & -1 \\ 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

We can also verify the exercise $A = BB^T - D$ directly:

$$BB^T = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix}.$$

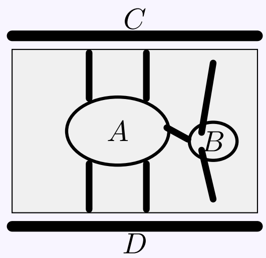
Hence

$$BB^T - D = \begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 3 & 1 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix} = A.$$

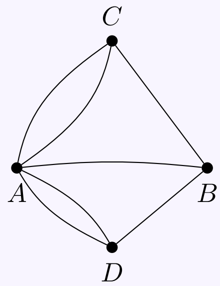
Definition 1.1.41 (k-walk)**.** *Let $G = (V, E)$ be a graph. A k-*walk *in G is a sequence of vertices $(v_0, v_1, \ldots, v_k)$ such that $\{v_i, v_{i+1}\} \in E$ for all $i = 0, 1, \ldots, k-1$. A walk* *is closed if $v_0 = v_k$. If a k-walk is a path, then all vertices are distinct. If a k-walk is a cycle, then all vertices are distinct except for the first and last vertices.*

第二章　Lecture 2

Topic 2.0.1 (The Seven Bridges of Königsberg)**.** *In Königsberg (now Kaliningrad), the river Pregel passed through the city with two islands and seven bridges connecting the islands and the two banks (see figure). The problem: can one start at some place, cross each bridge exactly once, and return to the start? Euler (1736) abstracted the two banks and two islands as four vertices and the seven bridges as seven edges, obtaining the graph G; he proved that such a closed trail exists if and only if every vertex has even degree. In G all four vertices have odd degree, so no solution exists. This problem is regarded as the origin of graph theory and Eulerian trails.*



Schematic of the seven bridges



Abstract graph G

Definition 2.0.2. *An Eulerian path is a walk which contains each edge of a graph exactly once.*

Definition 2.0.3. *An Eulerian cycle is a closed Eulerian path. A graph is called Eulerian if it contains an Eulerian cycle.*

Lemma 2.0.4. *Let $G = (V, E)$ be a graph with $|E| \geq 1$. If every vertex has degree at least 2 (i.e., $\deg(v) \geq 2$ for all $v \in V$), then G contains a cycle.*

证明. Take a maximal path $P = (v_0, v_1, \ldots, v_k)$ in G (so no edge extends P at either end). Since $|E| \geq 1$, we have $k \geq 1$. The vertex v_k has $\deg(v_k) \geq 2$, so v_k has a neighbor other than v_{k-1}. Any such neighbor must lie on P: if v_k were adjacent to some $u \notin \{v_0, \ldots, v_{k-1}\}$, then P could be extended to $v_0 \cdots v_k u$, contradicting maximality. So v_k is adjacent to v_i for some i with $0 \leq i \leq k-2$. Then $(v_i, v_{i+1}, \ldots, v_k, v_i)$ is a cycle in G. $\square$

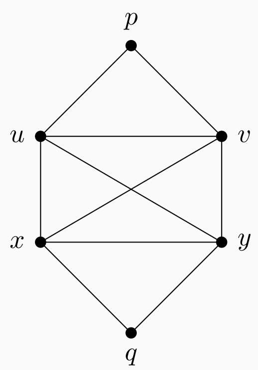
Theorem 2.0.5. *A finite connected graph is Eulerian if and only if every vertex has even degree.*

证明. *Necessity* ($\Rightarrow$). Let $C = (x_0, x_1, \dots, x_n)$ be an Eulerian cycle with $x_0 = x_n$. Each edge is used exactly once. For any vertex $y \in V$, each time C passes through y it uses one edge to enter and one to leave; so the number of edges incident to y used by C is twice the number of times y appears in $(x_1, \dots, x_{n-1})$. The start/end vertex $x_0 = x_n$ is left once and entered once, contributing 2 more. Thus $\deg(y)$ is even for all y.

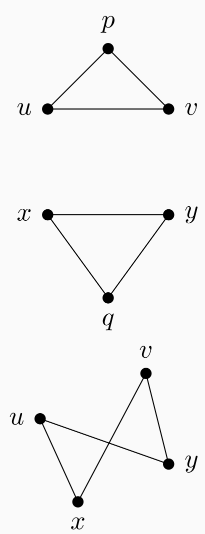
Sufficiency ($\Leftarrow$). (Hierholzer, 1873.) We use induction on $|E|$. If $|E| = 0$, then G is a single vertex and the trivial closed walk is an Eulerian cycle. Suppose $|E| \geq 1$. Since G is connected and every vertex has even degree, $\delta(G) \geq 2$, so G contains a cycle by the lemma above. Start at any vertex x and take a maximal walk that does not repeat any edge. Whenever the walk enters a vertex other than x, it has used an odd number of edges at that vertex so far, so an odd number (hence at least one) remains; thus the walk cannot get stuck except at x. So the walk returns to x, forming a closed trail C_1 with edge set $E_1 \subseteq E$.

Remove E_1 from G. In $G' = (V, E \setminus E_1)$, every vertex still has even degree (each vertex on C_1 lost 0 or 2 incident edges). So every non-trivial component of G' has an Eulerian cycle by the induction hypothesis. If $E_1 = E$, then C_1 is an Eulerian cycle. Otherwise, pick a vertex v that lies on C_1 and belongs to a component H of G' with at least one edge. That component has an Eulerian cycle C_2 (starting and ending at v). Splice C_2 into C_1 at v: traverse C_1 until v, then traverse all of C_2, then continue C_1 from v. Repeating this for every such component yields an Eulerian cycle of G. The algorithm runs in $O(|E|)$ time. □

Example 2.0.6. *Consider the graph below. On the left is the original connected graph; on the right we redraw the three edge-disjoint cycles that appear in the proof of Hierholzer's theorem.*



original graph



$C_1 = (u, x, v, y, u)$ *redrawn*

Choose the first closed trail to be

$$C_1 = (u, x, v, y, u),$$

which uses the four edges ux, xv, vy, *and* yu. *This is the closed trail: start at* u, *keep following unused edges, and because every time we enter a vertex we can also leave it again, the trail cannot get stuck anywhere except at the starting vertex. Hence the walk comes back to* u *and closes up.*
Now remove the edge set $E_1 = \{ux, xv, vy, yu\}$. *The remaining graph* $G' = (V, E \setminus E_1)$ *has exactly two non-trivial components: the top triangle* (u, p, v, u) *and the bottom triangle* (x, q, y, x). *Every vertex in these components still has even degree, so by the induction hypothesis each component has its own Eulerian cycle.*
Finally, splice those smaller cycles into C_1. *Start to traverse* C_1 *at* u, *but before leaving* u *for the edge* ux, *first run through the top triangle* (u, p, v, u). *When the tour later reaches* x, *run through the bottom triangle* (x, q, y, x), *and then continue along the original trail. One Eulerian cycle of the whole graph is therefore*

$$(u, p, v, u, x, q, y, x, v, y, u).$$

This illustrates the sentence "traverse C_1 *until* v, *then traverse all of* C_2, *then continue* C_1 *from* v*": each smaller Eulerian cycle is inserted at a shared vertex, and after all insertions every edge of the original graph has been used exactly once.*

Definition 2.0.7. *A Hamiltonion cycle is a closed walk which uses every vertex precisely once. A graph is called Hamiltonian if it contains a Hamiltonian cycle.*

Theorem 2.0.8. *Let* $G = (V, E)$ *be a simple graph on* $n \geq 3$ *vertices. If for every vertex* $v \in V$, $\deg(v) \geq \frac{n+1}{2}$, *then* G *is Hamiltonian.*

证明. First, we show that G is connected. Suppose for a contradiction that G is disconnected. Let C be the smallest connected component of G. The number of vertices in C is at most $\lfloor \frac{n}{2} \rfloor$. Thus, the maximum degree of any vertex in C is at most $\lfloor \frac{n}{2} \rfloor - 1 < \frac{n+1}{2}$, which contradicts the degree condition. Hence, G is connected.

We proceed by contradiction. Assume that G is not Hamiltonian. Let $P = v_1 v_2 \ldots v_k$ be a longest path in G. Since G is connected and $n \geq 3$, we have $k \geq 3$.

Because P is a longest path, all neighbors of its endpoints v_1 and v_k must lie within the set $\{v_1, v_2, \ldots, v_k\}$. If they had neighbors outside P, we could extend P to a longer path, contradicting its maximality.

Consider the sets of indices of the neighbors of v_1 and v_k on the path P:

$$S = \{i \in \{1, 2, \ldots, k-1\} \mid \{v_1, v_{i+1}\} \in E\}$$
$$T = \{i \in \{1, 2, \ldots, k-1\} \mid \{v_i, v_k\} \in E\}$$

Note that $|S| = \deg(v_1)$ and $|T| = \deg(v_k)$. By the given condition, we have:

$$|S| + |T| = \deg(v_1) + \deg(v_k) \geq \frac{n+1}{2} + \frac{n+1}{2} = n + 1$$

Since both S and T are subsets of $\{1, 2, \ldots, k-1\}$, and $k \leq n$, the maximum possible size of their union is $k - 1 \leq n - 1$.

By the Principle of Inclusion-Exclusion (or the Pigeonhole Principle), we have:

$$|S \cap T| = |S| + |T| - |S \cup T| \geq (n+1) - (n-1) = 2 > 0$$

Therefore, the intersection $S \cap T$ is non-empty. There exists some index i $(1 \leq i < k)$ such that $\{v_1, v_{i+1}\} \in E$ and $\{v_i, v_k\} \in E$.

Using these two edges, we can construct a cycle C containing exactly the vertices of P:

$$C = v_1 v_2 \ldots v_i v_k v_{k-1} \ldots v_{i+1} v_1$$

Since we assumed G is not Hamiltonian, C does not contain all vertices of G, so $k < n$. Because G is connected, there must exist a vertex $u \notin V(C)$ that is adjacent to some vertex $v_j \in V(C)$.

Since v_j is on the cycle C, we can break the cycle by removing one of the edges incident to v_j in C, say $\{v_{j-1}, v_j\}$. We then add the edge $\{u, v_j\}$ to form a new path:

$$P' = u v_j v_{j+1} \ldots v_k \ldots v_1 \ldots v_{j-1}$$

The length of P' is $k + 1$, which strictly exceeds the length of P. This contradicts the assumption that P is a longest path in G.

Thus, our assumption that G is not Hamiltonian must be false. G contains a Hamiltonian cycle. □

Remark 2.0.9. *The statement "Random graphs are Hamiltonian" asserts that a random graph $G(n, p)$ contains a Hamiltonian cycle asymptotically almost surely (a.a.s.) provided the edge probability p exceeds a sharp threshold. Specifically, if*

$$p \geq \frac{\ln n + \ln \ln n + \omega(1)}{n},$$

then

$$\lim_{n \to \infty} \mathbb{P}(G(n, p) \textit{ is Hamiltonian}) = 1.$$

This contrasts with the NP-completeness of the Hamiltonian cycle problem for general graphs.

The Hamiltonian cycle problem is a fundamental example of an **NP-complete** problem. In the context of computational complexity:

- **P (Polynomial time)** represents the class of problems that can be solved efficiently (in polynomial time) by a deterministic computer (Turing machine).
- **NP (Nondeterministic Polynomial time)** consists of problems where a solution,

once guessed, can be verified efficiently. Equivalently, these are solvable by a non-deterministic Turing machine in polynomial time.

- **NP-complete** problems are the "hardest" problems in NP. A problem is NP-complete if it is in NP and every other problem in NP can be transformed (reduced) into it in polynomial time.

If an efficient algorithm exists for any NP-complete problem, then efficient algorithms exist for all problems in NP (i.e., P = NP).

Graph theory is a fundamental branch of mathematics that studies the properties of graphs, which are mathematical structures used to model pairwise relations between objects. A graph consists of vertices (also called nodes or points) connected by edges (also called links or lines). The field has grown tremendously since Euler's solution to the Königsberg bridge problem in 1736, and now encompasses numerous subdisciplines:

- **Structural graph theory** investigates the intrinsic properties and structural characterizations of graphs. Central topics include graph decompositions, minors, and the celebrated Graph Minor Theorem of Robertson and Seymour. This area explores fundamental questions about graph embeddings, planarity, and forbidden subgraph characterizations.

- **Algorithmic graph theory** focuses on the design and analysis of algorithms for graph problems. It addresses computational complexity, approximation algorithms, and efficient data structures for graph representation. Classic problems include shortest paths, maximum flow, graph coloring, and graph isomorphism. The interplay between structural properties and algorithmic efficiency is a key theme.

- **Extremal graph theory** studies the extremal (maximum or minimum) values of graph invariants under given constraints. A prototypical question is: what is the maximum number of edges in a graph on n vertices that does not contain a specified subgraph? The foundational Turán's theorem and the Erdős–Stone theorem are cornerstones of this field.

- **Spectral graph theory** explores connections between graph properties and the eigenvalues and eigenvectors of associated matrices (adjacency matrix, Laplacian matrix). It provides powerful algebraic tools for studying graph connectivity, expansion properties, and random walks on graphs.

- **Random graph theory** investigates properties of graphs generated by random processes, such as the Erdős–Rényi model $G(n, p)$. It studies phase transitions, threshold phenomena, and the typical behavior of graph parameters.

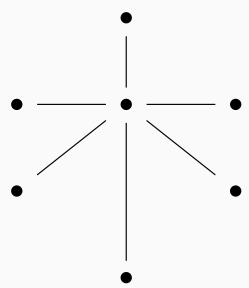
- **Algebraic graph theory** applies algebraic methods—including group theory, linear algebra, and representation theory—to study graphs. Topics include graph automorphisms, Cayley graphs, and association schemes.

These subfields are deeply interconnected and find applications across computer science, physics, biology, social network analysis, and numerous other disciplines.

2.1 Tree

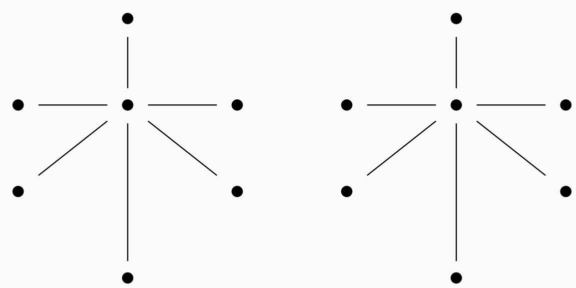
Definition 2.1.1. *A connected graph with no cycles is called a tree.*

Example 2.1.2. *This is an example of a tree. It is " 木 " in Chinese.*



Definition 2.1.3. *Disjoint union of trees is called a forest.*

Example 2.1.4. *This is an example of a forest. It is " 林 " in Chinese.*



Definition 2.1.5 (Spanning tree)**.** *Let $G = (V, E)$ be a connected (finite) graph. A* spanning tree *of G is a subgraph $T = (V, E_T)$ of G such that*

1. *$V(T) = V(G)$ (so T is spanning), and*
2. *T is a tree (equivalently: T is connected and contains no cycles).*

Equivalently, a spanning tree is a spanning, connected, acyclic subgraph of G (hence $|E_T| = |V| - 1$).

Proposition 2.1.6. *Every connected (finite) graph $G = (V, E)$ has a spanning tree.*

证明. Let $G = (V, E)$ be connected. Among all spanning subgraphs $H = (V, E_H)$ of G that are connected, choose one with $|E_H|$ minimal (such an H exists since E is finite and G itself is admissible). We claim that H is acyclic.

Indeed, if H contains a cycle C, pick an edge $e \in E(C)$ and let $H' := (V, E_H \setminus \{e\})$. Then H' is still connected, because the endpoints of e remain joined by the rest of the cycle, and any path using e can be rerouted along $C \setminus \{e\}$. This contradicts the minimality of

$|E_H|$.

Hence H is connected and acyclic, so H is a tree. Since $V(H) = V(G)$, H is a spanning tree of G. □

Theorem 2.1.7. *The followings are equivalent:*

1. *$G = (V, E)$ is a tree.*
2. *Any two vertices are connected by a unique path.*
3. *G is connected with $|V| - 1 = |E|$.*

证明. (1) $\implies$ (2) If two vertices x, y are connected by two distinct paths, then their union contains a cycle.

(2) $\implies$ (1) If G contains a cycle, then any two distinct vertices on the cycle are connected by at least two different paths. Connectivity is given.

(1) $\implies$ (3) We proceed by induction on $|V|$. The base case $|V| = 1$ is trivial. For $|V| > 1$, let $P = (u, u_1, \dots, u_l)$ be a longest path. Since G is a tree, all neighbors of the endpoint u must lie on P, which implies u is a leaf (degree 1). Removing this leaf and its incident edge leaves a tree on $|V| - 1$ vertices, and the result follows by induction.

(3) $\implies$ (1) Since G is connected, it has a spanning tree T. We know $|E(T)| = |V(T)| - 1 = |V(G)| - 1$. The given condition is $|E(G)| = |V(G)| - 1$. Since T is a subgraph of G, we must have $E(G) = E(T)$, which implies $G = T$. Thus, G is a tree.

□

Corollary 2.1.8. *Let G be a forest, then $\sharp$ connected components of $G = |V(G)| - |E(G)|$.*

证明. Let G be a forest, and let its connected components be

$$G_1, \dots, G_c,$$

where c is the number of connected components. Set

$$n_i := |V(G_i)|, \qquad m_i := |E(G_i)|.$$

Since G is a forest, each G_i is a tree. A basic fact about trees is that a tree on n_i vertices has exactly $n_i - 1$ edges, i.e.

$$m_i = n_i - 1 \qquad (1 \le i \le c).$$

Summing over all components gives

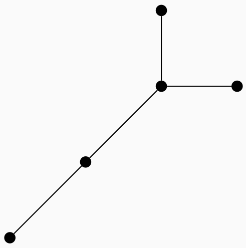
$$|E(G)| = \sum_{i=1}^{c} m_i = \sum_{i=1}^{c} (n_i - 1) = \sum_{i=1}^{c} n_i - c = |V(G)| - c.$$

Rearranging yields

$$c = |V(G)| - |E(G)|,$$

as claimed. □

Example 2.1.9. *Let T be a tree, $n := |V(T)| \geq 2$ with degree sequence $(d_1, \ldots, d_n)$, ordered by non-increasing. For example:*

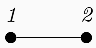


The degree sequence is $(3, 2, 1, 1, 1)$.

What is the number of trees on n vertices? $t(n)$. Cayley's formula solved this problem.

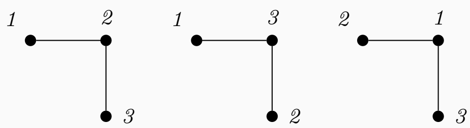
Example 2.1.10. *There are the basic examples:*

$n = 2$



$t(2) = 1$

$n = 3$



$t(3) = 3$

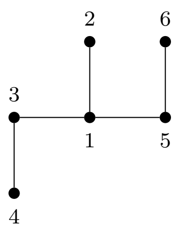
Theorem 2.1.11. $t(n) = n^{n-2}$.

证明. **Proof 1:** Goal: find a bijection between trees on sets $[n]$ and $[n]^{n-2}$.

Let T be a tree

1. Pick $u_1 \in u = \min\{v \in V : \deg(v) = 1\}$ and $a_1 :=$ the unique neighbor of u_1.
2. Remove u_1, write down the lable of a_1, and repeat.

For example:



We pick $u_1 = 2$, $a_1 = 1$, and remove u_1. Then we pick $u_2 = 4$, $a_2 = 3$, and remove u_2. Then we pick $u_3 = 3$, $a_3 = 1$, and remove u_3. Then we pick $u_4 = 1$, $a_4 = 5$, and remove u_4. Now we get a sequence $a = (1, 3, 1, 5)$ in $[6]^4$.

After all, we get a sequence $a = (a_1, \ldots, a_{n-2})$ in $[n]^{n-2}$. Next we want to show: For eah sequence a in $[n]^{n-2}$, there exists a unique tree T.

Let $\mathrm{d_i} := \sharp$ degree of vertex i and $f_i := \sharp$ occurrences of i in a.

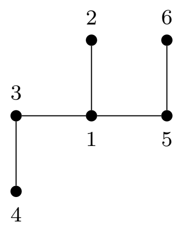
Note that $f_i \le d_i - 1$ for all i. Hence $n-2 = \sum f_i \le \sum(d_i - 1) = (2n-2) - n = n-2$. Hence $f_i = d_i - 1$ for all i.

Now we construct T:

1. For $i = 1, \ldots, n-2$: choose the smallest $b_i \in [n]$ not in $(a_i, a_{i+1}, \ldots, a_{n-2})$ and not equal to any b_j $(j < i)$; add edge (a_i, b_i).

2. So far we have $n-2$ edges. The two vertices that never appear as $b_1, \ldots, b_{n-2}$ are exactly the last "remaining" pair; add the edge between them to get the $(n-1)$-th edge. Thus we obtain a tree T.

For example: given $a = (1, 3, 1, 5)$, we can construct the tree step by step :



Hence, we have a bijection between trees on sets $[n]$ and $[n]^{n-2}$. □

证明. **Proof 2:** Let $d_1, \ldots, d_n$ be positive integers with $\sum_{i=1}^{n} d_i = 2(n-1)$, and let

$$t(d_1, \ldots, d_n) := \#\{\text{labeled trees on } [n] = \{1, \ldots, n\} \text{ with } \deg(i) = d_i\}.$$

We first prove the degree-sequence formula

$$t(d_1, \ldots, d_n) = \binom{n-2}{d_1 - 1, \ldots, d_n - 1} = \frac{(n-2)!}{\prod_{i=1}^{n}(d_i - 1)!}. \tag{$*$}$$

Step 1: a leaf-removal recurrence. Assume $d_n = 1$ (we may relabel so that one leaf is vertex n). In any such tree, the leaf n is adjacent to a unique vertex $i \in \{1, \ldots, n-1\}$. Removing n and its incident edge yields a labeled tree on $[n-1]$ whose degree sequence is

$$(d_1, \ldots, d_i - 1, \ldots, d_{n-1}).$$

Conversely, given a labeled tree on $[n-1]$ with degrees $(d_1, \ldots, d_i - 1, \ldots, d_{n-1})$, attaching the new leaf n to i produces a labeled tree on $[n]$ with degrees $(d_1, \ldots, d_{n-1}, 1)$. Hence

$$t(d_1, \ldots, d_{n-1}, 1) = \sum_{i=1}^{n-1} t(d_1, \ldots, d_i - 1, \ldots, d_{n-1}), \tag{R}$$

where terms with $d_i = 1$ are understood to contribute 0.
Step 2: verification for the multinomial expression. Define

$$F(d_1, \dots, d_n) := \frac{(n-2)!}{\prod_{j=1}^{n}(d_j - 1)!}.$$

For $d_n = 1$ we have

$$F(d_1, \dots, d_{n-1}, 1) = \frac{(n-2)!}{\prod_{j=1}^{n-1}(d_j - 1)!}.$$

Moreover, for $1 \le i \le n-1$,

$$F(d_1, \dots, d_i - 1, \dots, d_{n-1}) = \frac{(n-3)!}{(d_i - 2)! \prod_{j \ne i,\, j \le n-1}(d_j - 1)!} = \frac{(n-3)!(d_i - 1)}{\prod_{j=1}^{n-1}(d_j - 1)!}.$$

Summing over i gives

$$\sum_{i=1}^{n-1} F(d_1, \dots, d_i - 1, \dots, d_{n-1}) = \frac{(n-3)!}{\prod_{j=1}^{n-1}(d_j - 1)!} \sum_{i=1}^{n-1}(d_i - 1).$$

Since $\sum_{i=1}^{n-1} d_i = 2(n-1) - 1 = 2n - 3$, we have

$$\sum_{i=1}^{n-1}(d_i - 1) = (2n-3) - (n-1) = n - 2,$$

and therefore the right-hand side equals

$$\frac{(n-3)!(n-2)}{\prod_{j=1}^{n-1}(d_j - 1)!} = \frac{(n-2)!}{\prod_{j=1}^{n-1}(d_j - 1)!} = F(d_1, \dots, d_{n-1}, 1).$$

Thus F satisfies the same recurrence as t.
Step 3: induction. For $n = 2$ there is exactly one labeled tree, so $t(1,1) = 1 = F(1,1)$. By induction on n using the recurrence, we conclude $t(d_1, \dots, d_n) = F(d_1, \dots, d_n)$.
Step 4: summing over degree sequences. Let $t(n)$ be the total number of labeled trees on $[n]$. Summing over all degree sequences with $\sum d_i = 2(n-1)$ and setting $r_i = d_i - 1$ yields

$$t(n) = \sum_{\substack{r_1, \dots, r_n \ge 0 \\ r_1 + \cdots + r_n = n-2}} \binom{n-2}{r_1, \dots, r_n}.$$

By the multinomial theorem,

$$(x_1 + \cdots + x_n)^{n-2} = \sum_{r_1 + \cdots + r_n = n-2} \binom{n-2}{r_1, \dots, r_n} x_1^{r_1} \cdots x_n^{r_n}.$$

Evaluating at $x_1 = \cdots = x_n = 1$ gives

$$n^{n-2} = \sum_{r_1 + \cdots + r_n = n-2} \binom{n-2}{r_1, \dots, r_n} = t(n).$$

Hence $t(n) = n^{n-2}$, which is Cayley's formula. □

证明. **Proof 3:** Using Kirchhoff's Matrix–Tree Theorem.

Let $t(n)$ denote the number of spanning trees of the complete graph K_n. Write A for the adjacency matrix and D for the diagonal degree matrix. The (combinatorial) Laplacian is

$$L(K_n) = D - A.$$

In K_n every vertex has degree $n-1$, hence $D = (n-1)I$. Moreover $A = J - I$, where J is the all-ones matrix. Therefore

$$L(K_n) = (n-1)I - (J - I) = nI - J.$$

We determine the eigenvalues of $L(K_n)$. Let $\mathbf{1} = (1, \ldots, 1)^\mathsf{T}$. Since $J\mathbf{1} = n\mathbf{1}$, we have

$$L(K_n)\mathbf{1} = (nI - J)\mathbf{1} = n\mathbf{1} - n\mathbf{1} = 0,$$

so 0 is an eigenvalue (with eigenvector $\mathbf{1}$). If $x \in \mathbb{R}^n$ satisfies $\mathbf{1}^\mathsf{T} x = \sum_{i=1}^n x_i = 0$, then each entry of Jx equals $\sum_{i=1}^n x_i = 0$, hence $Jx = 0$, and thus

$$L(K_n)x = (nI - J)x = nx.$$

Consequently, n is an eigenvalue on the $(n-1)$-dimensional subspace $\mathbf{1}^\perp = \{x : \sum_i x_i = 0\}$, so the spectrum of $L(K_n)$ is

$$\underbrace{n, \ldots, n}_{n-1 \text{ times}}, \quad 0.$$

By Kirchhoff's Matrix–Tree Theorem (eigenvalue form), if the Laplacian eigenvalues are $0 = \lambda_n < \lambda_1 \le \cdots \le \lambda_{n-1}$, then the number of spanning trees is

$$\tau(G) = \frac{1}{n} \prod_{i=1}^{n-1} \lambda_i.$$

Applying this to $G = K_n$ gives

$$t(n) = \tau(K_n) = \frac{1}{n} \cdot n^{n-1} = n^{n-2}.$$

This is Cayley's formula. □

第三章 Lecture 3

3.1 Chromatic number

Definition 3.1.1. *Let $G = (V, E)$ be a graph and let C be a set of colors. A* proper coloring *of G is a map*

$$f : V \to C$$

such that for every edge $\{x, y\} \in E$ (equivalently, for every pair of adjacent vertices $x \sim y$) we have

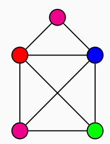
$$f(x) \neq f(y).$$

If G admits a proper coloring using at most k colors (that is, $|C| = k$), then G is called k-colorable.

The chromatic number *of G, denoted by $\chi(G)$, is the minimum number of colors needed for a proper coloring of G:*

$$\chi(G) = \min\{\, k : G \text{ is } k\text{-colorable}\}.$$

Example 3.1.2. *Consider following graph:*



We have $\chi(G) = 4$.

Corollary 3.1.3. *A graph G is bipartite if and only if $\chi(G) = 2$.*

证明. ($\Rightarrow$) Suppose G is bipartite. Then $V(G)$ can be written as a disjoint union

$$V(G) = A \cup B,$$

where every edge of G has one endpoint in A and the other in B. Define a coloring $f : V(G) \to \{1, 2\}$ by

$$f(v) = \begin{cases} 1, & v \in A, \\ 2, & v \in B. \end{cases}$$

If $\{u, v\} \in E(G)$, then u and v lie in different parts, hence $f(u) \neq f(v)$. Thus this is a proper coloring using two colors, so $\chi(G) \le 2$. Since G contains an edge, at least two colors are required, hence $\chi(G) = 2$.

($\Leftarrow$) Suppose $\chi(G) = 2$. Then there exists a proper coloring

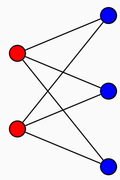
$$f : V(G) \to \{1, 2\}.$$

Let

$$A = f^{-1}(1), \qquad B = f^{-1}(2).$$

Then $V(G) = A \cup B$ and $A \cap B = \varnothing$. Since the coloring is proper, every edge $\{u, v\} \in E(G)$ satisfies $f(u) \ne f(v)$, so one endpoint lies in A and the other in B. Hence there are no edges inside A or inside B, and G is bipartite. □

Example 3.1.4. *Complete bipartite graph $K_{2,3}$:*



Theorem 3.1.5. *If G is planar then $\chi(G) \le 4$.*

Corollary 3.1.6. $\chi(K_n) = n$.

证明. In the complete graph K_n, every pair of distinct vertices is adjacent. Hence in any proper coloring, adjacent vertices must receive different colors, so all n vertices must have distinct colors. Thus $\chi(K_n) \ge n$.

On the other hand, assigning a different color to each vertex gives a proper coloring with n colors. Hence $\chi(K_n) \le n$.

Therefore $\chi(K_n) = n$. □

Corollary 3.1.7. *For a cycle C_n,*

$$\chi(C_n) = \begin{cases} 2, & \text{if } n \text{ is even,} \\ 3, & \text{if } n \text{ is odd.} \end{cases}$$

证明. Let $V(C_n) = \{v_1, \ldots, v_n\}$ where $v_i \sim v_{i+1}$ for $1 \le i < n$ and $v_n \sim v_1$.

If n is even, color the vertices alternately by two colors along the cycle:

$$f(v_i) = \begin{cases} 1, & i \text{ odd,} \\ 2, & i \text{ even.} \end{cases}$$

Since n is even, v_n and v_1 receive different colors, so this is a proper coloring. Hence $\chi(C_n) \le 2$. Since C_n has an edge, $\chi(C_n) \ge 2$, thus $\chi(C_n) = 2$.

If n is odd, alternating two colors along the cycle forces v_n and v_1 to have the same color, contradicting that they are adjacent. Hence $\chi(C_n) \geq 3$. Coloring the vertices alternately with two colors and giving v_n a third color yields a proper coloring with three colors, so $\chi(C_n) \leq 3$. Therefore $\chi(C_n) = 3$. □

Example 3.1.8. *consider the following graphs:*

Theorem 3.1.9 (Brooks' Theorem)**.** *Let $d \geq 3$ and let G be a graph with maximum degree d. If K_{d+1} is not a subgraph of G, then*

$$\chi(G) \leq d.$$

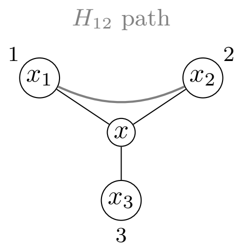
证明. Suppose, for a contradiction, that the theorem is false. Let G be a counterexample with the minimum number of vertices. Then $\chi(G) = d + 1$.

Let $x \in V(G)$ and let $N(x) = \{x_1, \ldots, x_\ell\}$, where $\ell \leq d$. Let $H = G - x$. By the minimality of G, we have $\chi(H) \leq d$. Take a proper d-coloring of H.

If fewer than d colors appear in $N(x)$, then we can color x with a color not used by its neighbors, obtaining a d-coloring of G, a contradiction. Hence $\ell = d$ and the vertices $x_1, \ldots, x_d$ receive pairwise distinct colors $1, \ldots, d$.

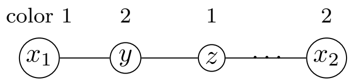
For $1 \leq i < j \leq d$, let H_{ij} be the subgraph of H induced by the vertices colored i or j. If x_i and x_j lie in different components of H_{ij}, then exchanging the colors i and j in the component containing x_i changes the color of x_i to j but leaves x_j unchanged. Thus color i disappears from $N(x)$, and we may color x with i, again obtaining a d-coloring of G, a contradiction.

Therefore, x_i and x_j lie in the same component of H_{ij} for all $i \neq j$. In particular, there exists an i–j path between them using only colors i and j.



If some vertex y in such a component had three neighbors in it, it would have at most $d - 3$ neighbors of other colors in H. Consequently, y is adjacent to at most $d - 2$ distinct colors in total. We can recolor y with a missing color $k \notin \{i, j\}$, which removes y from H_{ij}. Since x_i must have exactly one neighbor of color j in H_{ij} (otherwise its total degree in G would exceed d), the component of H_{ij} containing x_i and x_j must be a simple path (a Kempe chain); otherwise, recoloring a vertex of degree ≥ 3 would disconnect x_i and x_j, yielding the previous contradiction.

Since K_{d+1} is not a subgraph of G, the vertices in $N(x)$ cannot form a clique. Assume without loss of generality that x_1 and x_2 are not adjacent. The path H_{12} connecting x_1 and x_2 must contain at least one intermediate vertex. Let y be the neighbor of x_1 on H_{12}. The color of y is 2. Let z be the other neighbor of y on H_{12}, which has color 1.



path H_{12}

Now, consider the Kempe chain H_{13} connecting x_1 and x_3. We interchange the colors 1 and 3 on H_{13}. After this swap, x_1 receives color 3. The vertex y retains color 2, as it is not in H_{13}. To avoid a contradiction, x_1 and x_2 must still be connected by a new simple path H'_{23}. This path must start with the edge x_1y, meaning y must now have a second neighbor of color 3 to continue the path.

In the original coloring, y has exactly two neighbors of color 1 (x_1 and z) and exactly one neighbor of each remaining color $3, \ldots, d$. For y to gain a second neighbor of color 3 after the swap on H_{13}, its neighbor z must have belonged to H_{13} and changed its color from 1 to 3.

before: y's neighbors in H_{12} after swap on H_{13}

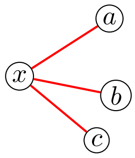
However, if $z \in H_{13}$, then z must have at least two neighbors of color 3 (to be in H_{13}), two neighbors of color 2 (to be in H_{12}), and at least one neighbor of each of the remaining $d-3$ colors (to prevent it from being easily recolored). This requires $\deg(z) \geq 2+2+(d-3) = d+1$, which contradicts $\max_{v \in V(G)} \deg(v) = d$.

This final contradiction demonstrates that no such minimal counterexample G can exist, completing the proof. $\square$

3.2 Ramsey theorem

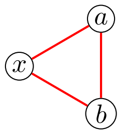
A classical motivation for Ramsey theory comes from the following social question posed in the 1950s: in any group of six people, there must exist either three people who all know each other or three people who are pairwise strangers.

This can be modeled using graph theory. Consider the complete graph K_6 whose vertices represent the six people. Color an edge red if the two corresponding people know each other and blue otherwise. Pick any vertex x. Among the five edges incident with x, at least three must have the same color (since $5 = 3 + 2$). Suppose, without loss of generality, that x has at least three red neighbors a, b, c.

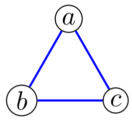


three red neighbors

If any pair among $\{a, b, c\}$ is joined by a red edge, say ab, then $\{x, a, b\}$ forms a red triangle. Otherwise the edges ab, bc, ca are all blue, and $\{a, b, c\}$ forms a blue triangle. Hence every red–blue coloring of the edges of K_6 contains a monochromatic triangle.



red triangle $\{x, a, b\}$



blue triangle $\{a, b, c\}$

This result is written as

$$R(3, 3) = 6,$$

where $R(s, t)$ denotes the smallest integer n such that every red–blue coloring of the edges of K_n contains either a red K_s or a blue K_t. The study of such unavoidable patterns in sufficiently large structures forms the basis of *Ramsey theory*.

$R(s, t)$ is the smallest integer n such that all edge colorings with 2 colors $\{c_1, c_2\}$ of K_n contain a monochromatic K_s in color c_1 or a monochromatic K_t in color c_2= smallest integer such that all graphs G on n vertices satisfies

(1) K_s is a subgraph of G or

(2) K_t is subgraph of $\bar{G}$

Corollary 3.2.1. *$R(2, t) = R(t, 2) = t$ for all $t \geq 2$.*

证明. We prove $R(2, t) = t$; the equality $R(t, 2) = t$ follows by symmetry.

Upper bound. In any red–blue coloring of $E(K_t)$, either there is a red edge, which is a red K_2, or else there are no red edges, in which case all edges are blue and the whole vertex set forms a blue K_t. Hence $R(2, t) \leq t$.

Lower bound. Consider K_{t-1} with all edges colored blue. Then there is no red K_2, and also no blue K_t (since there are only $t - 1$ vertices). Thus $R(2, t) \geq t$.

Therefore $R(2, t) = t$, and hence $R(t, 2) = t$. □

Proposition 3.2.2. $R(s, t) \leq R(s - 1, t) + R(s, t - 1)$

证明. Let

$$n := R(s - 1, t) + R(s, t - 1),$$

and consider any red–blue coloring of the edges of K_n. Fix a red vertex x. Let A be the set of red neighbors of x and B the set of blue neighbors of x. Then $|A| + |B| = n - 1$.

If $|A| \geq R(s-1,t)$, then the induced subgraph on A contains either a red K_{s-1} or a blue K_t. In the first case, together with x this forms a red K_s; in the second case we already have a blue K_t.

If $|A| < R(s-1,t)$, then

$$|B| = n - 1 - |A| \geq n - 1 - (R(s-1,t) - 1) = R(s,t-1).$$

as $n = R(s-1,t) + R(s,t-1)$ Hence the induced subgraph on B contains either a red K_s or a blue K_{t-1}. In the first case we already have a red K_s; in the second case, together with x this forms a blue K_t.

Thus every coloring of K_n contains a red K_s or a blue K_t, so $R(s,t) \leq n$. □

Corollary 3.2.3. $R(r,s) \leq \binom{r+s-2}{r-1}$.

证明. By induction on $r+s$. The base case $r+s=3$ is trivial. For the inductive step, assume the inequality holds for all $r+s' < r+s$. Then

$$R(r,s) \leq R(r-1,s) + R(r,s-1) \leq \binom{r+s-3}{r-2} + \binom{r+s-3}{s-2}.$$

By the inductive hypothesis,

$$\binom{r+s-3}{r-2} \leq \binom{r+s-2}{r-1} \quad \text{and} \quad \binom{r+s-3}{s-2} \leq \binom{r+s-2}{s-1}.$$

Hence $R(r,s) \leq \binom{r+s-2}{r-1}$. □

We first establish a recursive upper bound for the general Ramsey number $R(r,s)$, from which the diagonal bound follows.

Theorem 3.2.4 (Erdős-Szekeres Upper Bound)**.** *For any integers $s \geq 2$, the diagonal Ramsey number satisfies*

$$R(s,s) \leq \binom{2s-2}{s-1}.$$

Consequently, as $s \to \infty$, we have the asymptotic bound

$$R(s,s) \leq (1+o(1))\frac{4^{s-1}}{\sqrt{\pi s}}.$$

证明. The first part follows from the proof of the general bound $R(r,s) \leq \binom{r+s-2}{r-1}$.

Asymptotic Analysis: To derive the asymptotic behavior, we apply Stirling's approximation, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$:

$$\binom{2s-2}{s-1} = \frac{(2s-2)!}{((s-1)!)^2} \sim \frac{\sqrt{2\pi(2s-2)}\left(\frac{2s-2}{e}\right)^{2s-2}}{2\pi(s-1)\left(\frac{s-1}{e}\right)^{2s-2}} = \frac{4^{s-1}}{\sqrt{\pi(s-1)}}.$$

As $s \to \infty$, $\frac{1}{\sqrt{s-1}} = \frac{1}{\sqrt{s}}(1+o(1))$, yielding the desired asymptotic form:

$$R(s,s) \le (1+o(1))\frac{4^{s-1}}{\sqrt{\pi s}}.$$

□

This following theorem introduces the probabilistic method to combinatorics, demonstrating the existence of specific graphs by showing that the probability of their existence is strictly positive.

Theorem 3.2.5 (Erdős Exponential Lower Bound). *For any integer $n \ge s \ge 2$, if*

$$\binom{n}{s} 2^{1-\binom{s}{2}} < 1,$$

then $R(s,s) > n$. Consequently, we have the asymptotic lower bound:

$$R(s,s) \ge (1+o(1))\frac{s}{e\sqrt{2}} 2^{s/2}.$$

证明. Consider a complete graph K_n. We assign colors (red or blue) to each edge of K_n independently and uniformly at random with probability $\frac{1}{2}$.

Let S be a fixed subset of $V(K_n)$ of size s. Let A_S denote the event that the complete subgraph induced by S (i.e., K_s) is monochromatic (either all red or all blue). The probability of A_S occurring is:

$$\mathbb{P}(A_S) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{s}{2}} = 2^{1-\binom{s}{2}}.$$

Let X be the random variable counting the total number of monochromatic K_s subgraphs in K_n. We can express X as the sum of indicator variables: $X = \sum_{|S|=s} \mathbf{1}_{A_S}$. By the linearity of expectation, the expected number of monochromatic s-cliques is:

$$\mathbb{E}[X] = \sum_{|S|=s} \mathbb{P}(A_S) = \binom{n}{s} 2^{1-\binom{s}{2}}.$$

If $\mathbb{E}[X] < 1$, then by the properties of random variables, there must exist at least one point in the sample space (i.e., at least one specific 2-coloring of K_n) for which $X = 0$. Such a coloring contains strictly zero monochromatic K_s subgraphs. By definition, this implies that the Ramsey number $R(s,s)$ must be strictly greater than n.

Asymptotic Analysis: We seek the largest n such that the strict inequality holds. Using the standard upper bound for binomial coefficients $\binom{n}{s} \le \frac{n^s}{s!}$, it is sufficient to satisfy:

$$\frac{n^s}{s!} 2^{1-s(s-1)/2} < 1 \implies n^s < s! \cdot 2^{s(s-1)/2-1}.$$

Taking the s-th root of both sides gives:

$$n < (s!)^{1/s} \cdot 2^{(s-1)/2} \cdot 2^{-1/s}.$$

Applying Stirling's approximation $(s!)^{1/s} \sim \frac{s}{e}(\sqrt{2\pi s})^{1/s} \sim \frac{s}{e}$, and noting that $2^{(s-1)/2} = \frac{1}{\sqrt{2}}2^{s/2}$, we obtain:

$$n \approx \frac{s}{e} \cdot \frac{1}{\sqrt{2}}2^{s/2} = \frac{s}{e\sqrt{2}}2^{s/2}.$$

Therefore, setting n to be slightly less than this value guarantees the existence of a valid coloring, establishing the asymptotic lower bound:

$$R(s,s) \geq (1 + o(1))\frac{s}{e\sqrt{2}}2^{s/2}.$$

□

Historical and Recent Progress

Result Type	**Year**	**Researchers**	**Key Breakthrough**
Classic Upper Bound	1935	Erdős, Szekeres	$R(s,s) \leq D \cdot 4^{s-1}$ (based on combinatorics).
Upper Bound Impr.	2009	David Conlon	Sub-exponential improvement: $R(s,s) \leq s^{-c\frac{\log s}{\log\log s}}4^s$.
Upper Bound Brkthr.	2023/24	Campos, Griffiths, Morris, et al.	First exponential improvement: $R(s,s) \leq (4-\epsilon)^s$ ($\epsilon \approx 2^{-7}$).
Classic Lower Bound	1947	Paul Erdős	$R(s,s) > C \cdot 2^{s/2}$ (dawn of the probabilistic method).
Lower Bound Brkthr.	2025	Jie Ma, W. Shen, S. Xie	Improvement for the off-diagonal Ramsey numbers $R(s,Cs)$.

For off-diagonal Ramsey numbers, we introduce the following results:

Result Type	**Year**	**Researchers**	**Key Contribution**
Classic Upper Bound	1935	Erdős, Szekeres	$r(s,t) = O(t^{s-1})$ for all fixed $s \geq 3$ and $t \to \infty$.
Upper Bound Impr.	1980	Ajtai, Komlós, Szemerédi	First improvement to the upper bound on $r(s,t)$ by analyzing a randomized greedy algorithm for producing large independent sets.
Lower Bound	2010	Bohman, Keevash	Lower bound by analyzing the random K_s-free graph process, improving on earlier results of Spencer.
Upper Bound Impr.	2001	Li, Rousseau, Zang	Extending ideas of Shearer, showed that $r(s,t) \leq (1+o(1))\frac{t^{s-1}}{(\log t)^{s-2}}$ as $t \to \infty$.
Lower Bound Brkthr.	1995	Kim	$r(3,t) = \Omega(t^2/\log t)$, matching previous upper bounds by Ajtai, Komlós and Szemerédi and Shearer, and improving earlier bounds of Spencer.
Asymptotic Result	2020	Fiz Pontiveros, Griffiths, Morris; Bohman, Keevash	$r(3,t)$ is determined asymptotically up to a factor of four.

第四章　Lecture 4

4.1 Ramsey Theory on Structures and Spaces

Theorem 4.1.1 (Schur's Theorem). *Let n be a positive integer, and let s_n be the smallest integer such that every n-coloring of $[s_n] = \{1, 2, \ldots, s_n\}$ contains a monochromatic solution to*

$$x + y = z.$$

Then such an integer s_n exists.

证明. Identify the set of n colors with $[n] = \{1, 2, \ldots, n\}$, and let

$$N := R(\underbrace{3, \ldots, 3}_{n \text{ times}})$$

be the n-color Ramsey number for triangles.

We claim that every coloring of $[N]$ with n colors contains a monochromatic solution to $x + y = z$. This will imply that the set of positive integers with the required property is nonempty, and hence, by well-ordering, its least element s_n exists.

So let $c : [N] \to [n]$ be any coloring. Consider the complete graph K_N with vertex set $[N]$. Color each edge $\{u, v\}$, where $u \neq v$, by the color

$$c(|u - v|).$$

Since $N = R(3, \ldots, 3)$, Ramsey's theorem guarantees a monochromatic triangle in this edge-colored complete graph. Thus there exist distinct vertices

$$u, v, w \in [N]$$

such that the three edges $\{u, v\}$, $\{u, w\}$, and $\{v, w\}$ all have the same color.

Without loss of generality, assume

$$u > v > w.$$

Set

$$x = u - v, \qquad y = v - w, \qquad z = u - w.$$

Then $x, y, z \in [N]$, and

$$x + y = (u - v) + (v - w) = u - w = z.$$

Moreover, the numbers x, y, z all receive the same color under c, because

$$c(x) = c(|u - v|), \qquad c(y) = c(|v - w|), \qquad c(z) = c(|u - w|),$$

and these three edge colors are equal by construction of the monochromatic triangle.

Hence we have found a monochromatic solution to $x+y = z$. Therefore every n-coloring of $[N]$ contains such a solution, so the desired integer exists. □

Definition 4.1.2. *Define* $[q]^n := \{(x_1, \cdots, x_n) : x_i \in [q]\}$. *We say* $L \subseteq [q]^n$ *is a* combinatorial line *if there exists* $I \subseteq [n]$ *and integers* a_i *such that*

$$L = \{(x_1, \cdots, x_n) \in [q]^n : x_i = a_i \text{ for } i \notin I \text{ and } x_i = x_j \text{ for all } i, j \in I\}.$$

Example 4.1.3. *Consider* $[3]^3 = \{(x_1, x_2, x_3) : x_i \in \{1, 2, 3\}\}$. *Let*

$$I = \{1, 2\}, \qquad a_3 = 2.$$

Then the set

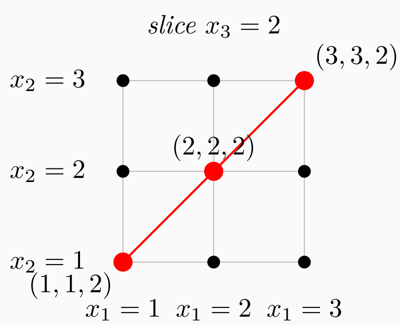
$$L = \{(x_1, x_2, x_3) \in [3]^3 : x_3 = 2 \text{ and } x_1 = x_2\}$$

is a combinatorial line. Explicitly,

$$L = \{(1, 1, 2), (2, 2, 2), (3, 3, 2)\}.$$

Indeed, the third coordinate is fixed, while the first two coordinates vary together and always take the same value.

A schematic picture is given below. Since the three points lie in the plane $x_3 = 2$, *we draw that slice as a* 3×3 *grid and highlight the diagonal corresponding to the line* $x_1 = x_2$.



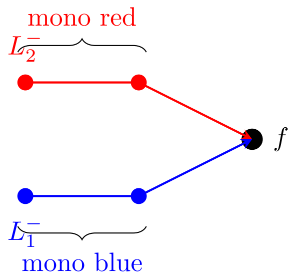
Theorem 4.1.4 (Hales–Jewett). *For all $r, q \in \mathbb{N}_+$, there exists a smallest $n := HJ(r, q)$ such that any r-coloring of $[q]^n$ contains a monochromatic combinatorial line.*

证明. The proof proceeds by induction on q. The case $q = 1$ is trivial (every point is a monochromatic line), so we assume $q \geqslant 2$ and that $HJ(r, q')$ is finite for all $q' \leqslant q - 1$.

Step 0: Key definitions. For a combinatorial line L in $[q]^n$, write L^- for its *first point* (all active coordinates equal 1) and L^+ for its *last point* (all active coordinates equal q).

We say that lines $L_1, \cdots, L_s$ are *focused at f* if $L_i^+ = f$ for every $i \in [s]$. They are *color-focused at f* (with respect to a coloring χ) if, additionally, each truncated line $L_i \setminus \{L_i^+\}$ is monochromatic and these s colors are pairwise different.

The following picture illustrates 2 lines color-focused at the common endpoint f (with $q = 3$, so each line has 3 points and the truncation removes the last one):



$L_1 \setminus \{f\}$ and $L_2 \setminus \{f\}$: different colors

Step 1: Reduction to a "focused" statement. We prove the following strengthening by induction on s: for each $s \leqslant r$, there exists $N = \mathrm{FHJ}(r, s, q)$ such that any r-coloring of $[q]^N$ either

(a) contains a monochromatic combinatorial line, or

(b) contains s color-focused lines at some common point.

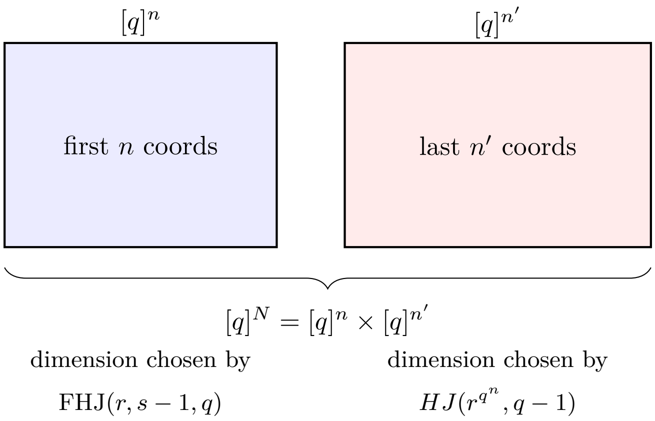
Taking $s = r$: in case (b), the r truncated lines use r pairwise different colors from a palette of r colors, so all r colors appear. At the common focus point f, the color of f must equal one of them, yielding a monochromatic line. Thus case (b) forces case (a).

Step 2: Base case ($s = 1$). Set $\mathrm{FHJ}(r, 1, q) = HJ(r, q - 1)$. By induction on q, any r-coloring of $[q-1]^{HJ(r,q-1)}$ contains a monochromatic line L (using values in $[q-1]$). This L already lies inside $[q]^{HJ(r,q-1)}$, and the truncation $L \setminus \{L^+\}$ is monochromatic, so $\{L\}$ is a single color-focused line at L^+.

Step 3: Inductive step. Suppose $\mathrm{FHJ}(r, s', q)$ is finite for all $s' \leqslant s - 1$. Set

$$n := \mathrm{FHJ}(r, s-1, q), \qquad n' := HJ(r^{q^n}, q-1), \qquad N := n + n'.$$

We show $\mathrm{FHJ}(r, s, q) \leqslant N$. The argument has three sub-steps, illustrated schematically below.



Step 3a: Splitting and super-coloring. Given an r-coloring χ of $[q]^N$, write each point as (a, b) where $a \in [q]^n$ (first n coordinates) and $b \in [q]^{n'}$ (last n' coordinates). Define a *super-coloring* χ' on $[q]^{n'}$ by assigning to each b the entire function

$$\chi'(b) := f_b, \qquad \text{where} \quad f_b \colon [q]^n \to [r], \quad a \mapsto \chi(a, b).$$

There are at most r^{q^n} possible such functions, so χ' is an r^{q^n}-coloring.

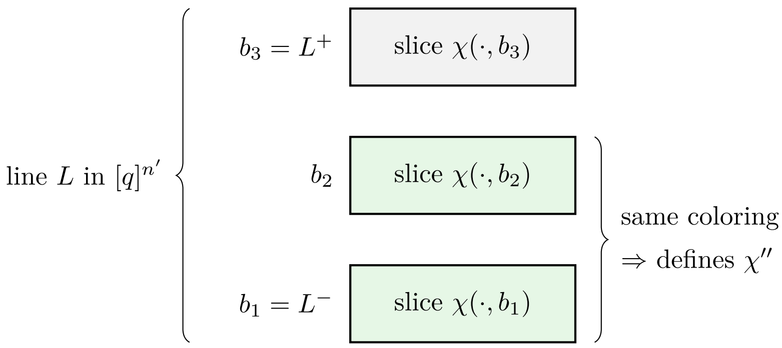
Intuition. Two points b, b' in $[q]^{n'}$ receive the same super-color if and only if the "slices" $[q]^n \times \{b\}$ and $[q]^n \times \{b'\}$ are colored identically under χ.

Step 3b: Applying Hales–Jewett in the second block. Since $n' = HJ(r^{q^n}, q-1)$, the super-coloring χ' on $[q-1]^{n'} \subseteq [q]^{n'}$ contains a monochromatic combinatorial line L with active coordinates $I' \subseteq \{n+1, \cdots, n+n'\}$. "Monochromatic under χ'" on $L \setminus \{L^+\}$ means:

$$\forall\, b, b' \in L \setminus \{L^+\} : \quad f_b = f_{b'} \quad (\text{identical coloring on } [q]^n).$$

So we may define a *well-defined* r-coloring on $[q]^n$:

$$\chi''(a) := \chi(a, b) \quad \text{for any } b \in L \setminus \{L^+\}.$$



Step 3c: Applying the focused induction in the first block. Since $n = \mathrm{FHJ}(r, s-1, q)$, the coloring χ'' on $[q]^n$ either

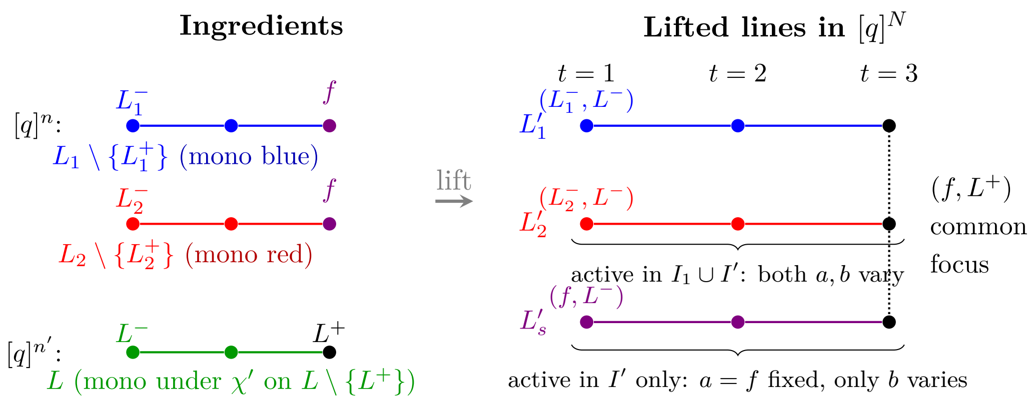
- contains a monochromatic line (done!), or
- contains $s-1$ lines $L_1, \cdots, L_{s-1}$ in $[q]^n$ that are color-focused at some point f.

Assume the latter. We now "lift" these $s-1$ lines to $[q]^N$ and add one more line to get s color-focused lines.

Step 4: Constructing s color-focused lines in $[q]^N$. Write $I_i \subseteq \{1, \cdots, n\}$ for the active coordinates of L_i in $[q]^n$. Define s lines in $[q]^N$:

- For $1 \leqslant i \leqslant s-1$: let L_i' be the line with first point (L_i^-, L^-) and active coordinates $I_i \cup I'$.
- For $i = s$: let L_s' be the line with first point (f, L^-) and active coordinates I' only.

We illustrate with $q = 3$, $s = 3$ (so $s-1 = 2$ old lines plus one new line). Each line has 3 points; writing them from first ($t = 1$) to last ($t = 3$):



Verification. All s lines share the common last point (f, L^+), so they are focused there. It remains to check that the truncations use pairwise different colors:

- For $1 \leqslant i \leqslant s-1$: the points of $L_i' \setminus \{(f, L^+)\}$ have the form (x, b) where x ranges over $L_i \setminus \{f\}$ and b ranges over $L \setminus \{L^+\}$. Since $f_b = f_{b'}$ for all $b, b' \in L \setminus \{L^+\}$, the color $\chi(x, b) = \chi''(x)$, which is the monochromatic color of $L_i \setminus \{f\}$ under χ''.
- For $i = s$: the points of $L_s' \setminus \{(f, L^+)\}$ have the form (f, b) where $b \in L \setminus \{L^+\}$. Their color is $\chi(f, b) = \chi''(f)$, which is the color of the focus f under χ''.

Since $L_1, \cdots, L_{s-1}$ were color-focused at f under χ'', their $s-1$ truncation colors are pairwise different and all differ from $\chi''(f)$, giving s pairwise different colors in total. Hence $L_1', \cdots, L_s'$ are color-focused at (f, L^+) under χ. □

4.2 Turán-type Problems I

Definition 4.2.1. *Let n be an integer, and let H be a graph. The* Turán number $\mathrm{ex}(n, H)$ *denotes the maximal number of edges in an H-free graph on n vertices. Here H-free means H is not a subgraph.*
More generally, define $\mathrm{ex}(n, \{H_1, \cdots, H_k\})$ for all H_i forbidden.

Lemma 4.2.2. $\mathrm{ex}(n, \{C_3, C_4, \cdots, C_n\}) = n - 1$.

Proposition 4.2.3 (Mantel, 1907)**.** $\mathrm{ex}(n, C_3) = \lfloor n^2/4 \rfloor$.

Proof 1. Let $G = (V, E)$ be C_3-free. Define $n = |V|$ and $m = |E|$. For $\{x, y\} \in E$, as G is C_3-free, $d(x) + d(y) \leqslant n$. By double-counting walks of length 2,

$$\sum_{x \in V} d(x)^2 = \sum_{\{x,y\} \in E} (d(x) + d(y)) \leqslant mn.$$

By Cauchy–Schwarz and $\sum d(x) = 2m$,

$$mn \geqslant \sum_{x \in V} d(x)^2 \geqslant \frac{(\sum d(x))^2}{n} = \frac{4m^2}{n}.$$

Hence $m \leqslant n^2/4$. □

Proof 2. Let A be the largest coclique (i.e. independent set) of G. As $N(x)$ is edge-less, $|A| \geqslant |N(x)| = d(x)$ for all $x \in V$. Also, as A is largest, $B := V \setminus A$ meets every edge. Hence $\sum_{x \in B} d(x) \geqslant |E|$. Therefore

$$|E| \leqslant \sum_{x \in B} d(x) \leqslant |B| \cdot |A| \leqslant \left(\frac{|A| + |B|}{2} \right)^2 = \frac{n^2}{4}.$$

□

Proof 3. We prove by induction on $m := |V(G)|$ that every triangle-free graph G on m vertices satisfies

$$|E(G)| \le \left\lfloor \frac{m^2}{4} \right\rfloor.$$

If $m \le 2$, then clearly $|E(G)| \le 1$, so the claim is immediate.

Now assume $m \ge 3$ and that the statement holds for all triangle-free graphs on $m - 2$ vertices. Let G be a triangle-free graph on m vertices.

If $E(G) = \varnothing$, then there is nothing to prove. So choose an edge $xy \in E(G)$, and let

$$H := G[V(G) \setminus \{x, y\}]$$

be the induced subgraph obtained by deleting x and y. Then H is triangle-free and has $m - 2$ vertices. Hence, by the induction hypothesis,

$$|E(H)| \le \left\lfloor \frac{(m-2)^2}{4} \right\rfloor.$$

Since G is triangle-free and $xy \in E(G)$, the vertices x and y have no common neighbor other than each other. Indeed, if some vertex $z \neq x, y$ were adjacent to both x and y, then x, y, z would form a triangle. Therefore

$$\big(N(x) \setminus \{y\}\big) \cap \big(N(y) \setminus \{x\}\big) = \varnothing.$$

Both sets are contained in $V(G) \setminus \{x, y\}$, which has size $m-2$. Thus

$$(d(x)-1)+(d(y)-1) \leq m-2,$$

and hence

$$d(x)+d(y) \leq m.$$

Every edge of G is either an edge of H, or is incident with x or y. Therefore

$$|E(G)| = |E(H)| + d(x) + d(y) - 1,$$

where we subtract 1 because the edge xy is counted in both $d(x)$ and $d(y)$. It follows that

$$|E(G)| \leq \left\lfloor \frac{(m-2)^2}{4} \right\rfloor + m - 1.$$

It remains to check that

$$\left\lfloor \frac{(m-2)^2}{4} \right\rfloor + m - 1 = \left\lfloor \frac{m^2}{4} \right\rfloor.$$

If $m = 2n$, then

$$\left\lfloor \frac{(m-2)^2}{4} \right\rfloor + m - 1 = \frac{(2n-2)^2}{4} + 2n - 1 = (n-1)^2 + 2n - 1 = n^2 = \left\lfloor \frac{(2n)^2}{4} \right\rfloor.$$

If $m = 2n+1$, then

$$\left\lfloor \frac{(m-2)^2}{4} \right\rfloor + m - 1 = \left\lfloor \frac{(2n-1)^2}{4} \right\rfloor + 2n = (n^2 - n) + 2n = n^2 + n = \left\lfloor \frac{(2n+1)^2}{4} \right\rfloor.$$

Thus in all cases,

$$|E(G)| \leq \left\lfloor \frac{m^2}{4} \right\rfloor.$$

Therefore

$$\mathrm{ex}(n, C_3) = \left\lfloor \frac{n^2}{4} \right\rfloor.$$

□

Theorem 4.2.4 (Turán, 1941)**.** *For every integer $k \geq 2$,*

$$\mathrm{ex}(n, K_{k+1}) = \left\lfloor \left(1 - \frac{1}{k}\right) \frac{n^2}{2} \right\rfloor.$$

Moreover, equality holds if and only if the graph is the complete k-partite graph whose part sizes differ by at most 1, i.e. the Turán graph

$$T_{n,k} = K_{\lceil n/k \rceil, \ldots, \lfloor n/k \rfloor}.$$

证明. Write

$$t_{n,k} := \left\lfloor \left(1-\frac{1}{k}\right)\frac{n^2}{2} \right\rfloor.$$

We prove simultaneously the upper bound and the uniqueness statement by induction on n, with a subsidiary induction on k.

The statement is trivial when $n \le k$, since then every graph on n vertices is K_{k+1}-free and

$$\mathrm{ex}(n, K_{k+1}) = \binom{n}{2} = t_{n,k}.$$

The case $k = 2$ is Mantel's theorem.

Now let $k \ge 3$ and $n \ge k+1$, and assume the theorem holds for all smaller values of n, and also for the parameter $k-1$.

Let G be a K_{k+1}-free graph on n vertices.

Step 1: the upper bound. If G contains no copy of K_k, then G is K_k-free, so by the induction hypothesis on $k-1$,

$$e(G) \le t_{n,k-1} < t_{n,k},$$

and hence in this case there is nothing more to prove.

Therefore we may assume that G contains a k-clique. Choose such a clique

$$A = \{a_1, \ldots, a_k\}, \qquad B := V(G) \setminus A.$$

Then

$$e(G) = e(A) + e(B) + e(A, B).$$

Since A is a k-clique,

$$e(A) = \binom{k}{2}.$$

Also, the induced subgraph $G[B]$ is still K_{k+1}-free, so by the induction hypothesis on $n-k < n$,

$$e(B) \le t_{n-k,k}.$$

Finally, every vertex $b \in B$ is adjacent to at most $k-1$ vertices of A, for otherwise $A \cup \{b\}$ would span a K_{k+1}. Hence

$$e(A, B) \le (k-1)(n-k).$$

Combining these estimates gives

$$e(G) \le \binom{k}{2} + t_{n-k,k} + (k-1)(n-k).$$

Since the first and third terms are integers,

$$\binom{k}{2} + t_{n-k,k} + (k-1)(n-k) \le \left\lfloor \binom{k}{2} + \left(1-\frac{1}{k}\right)\frac{(n-k)^2}{2} + (k-1)(n-k) \right\rfloor.$$

A direct calculation shows that

$$\binom{k}{2} + \left(1 - \frac{1}{k}\right)\frac{(n-k)^2}{2} + (k-1)(n-k) = \left(1 - \frac{1}{k}\right)\frac{n^2}{2},$$

therefore

$$e(G) \le t_{n,k}.$$

Thus

$$\mathrm{ex}(n, K_{k+1}) \le t_{n,k}.$$

Step 2: the extremal construction. The complete k-partite graph $T_{n,k}$ has no $(k+1)$-clique, since any clique in a k-partite graph contains at most one vertex from each part. Its number of edges is

$$e(T_{n,k}) = t_{n,k}.$$

Hence

$$\mathrm{ex}(n, K_{k+1}) \ge t_{n,k}.$$

Together with the upper bound, this yields

$$\mathrm{ex}(n, K_{k+1}) = t_{n,k}.$$

Step 3: the equality case. Suppose now that $e(G) = t_{n,k}$.

As observed above, G must contain a K_k, for otherwise $e(G) \le t_{n,k-1} < t_{n,k}$. Fix a k-clique

$$A = \{a_1, \dots, a_k\}, \qquad B := V(G) \setminus A.$$

Since equality holds in

$$e(G) \le \binom{k}{2} + e(B) + e(A,B) \le \binom{k}{2} + t_{n-k,k} + (k-1)(n-k),$$

we must have equality in each of the intermediate estimates. In particular:

(i) $e(B) = t_{n-k,k}$, so by the induction hypothesis on n, the graph $G[B]$ is the Turán graph $T_{n-k,k}$;

(ii) every vertex $b \in B$ is adjacent to exactly $k-1$ vertices of A.

By (ii), for each $b \in B$ there is a unique index $i(b) \in \{1, \dots, k\}$ such that b is *not* adjacent to $a_{i(b)}$. For each $i \in \{1, \dots, k\}$ define

$$C_i := \{a_i\} \cup \{\, b \in B : i(b) = i \,\}.$$

Then the sets $C_1, \dots, C_k$ form a partition of $V(G)$.

We claim that each C_i is an independent set. Indeed, by definition a_i is adjacent to no vertex of $C_i \setminus \{a_i\}$. If $x, y \in C_i \setminus \{a_i\}$ were adjacent, then both x and y would be adjacent to every vertex of $A \setminus \{a_i\}$; hence

$$\{x, y\} \cup (A \setminus \{a_i\})$$

would span a K_{k+1}, a contradiction. Thus each C_i is independent.

Therefore G is a spanning subgraph of the complete k-partite graph with $C_1, \dots, C_k$ parts, and so

$$e(G) \leq \sum_{1 \leq i < j \leq k} |C_i||C_j|.$$

For fixed $\sum_i |C_i| = n$, the quantity on the right is maximized exactly when the part sizes differ by at most 1, and its maximum is $t_{n,k}$. Since $e(G) = t_{n,k}$, we obtain

$$t_{n,k} = e(G) \leq \sum_{i<j} |C_i||C_j| \leq t_{n,k}.$$

Hence equality holds throughout. It follows that:

- the sizes $|C_1|, \dots, |C_k|$ differ by at most 1;
- all edges between distinct parts C_i and C_j are present.

Thus G is exactly the complete k-partite graph with part sizes as equal as possible, namely

$$G \cong T_{n,k} = K_{\lceil n/k \rceil, \dots, \lfloor n/k \rfloor}.$$

Conversely, $T_{n,k}$ is K_{k+1}-free and has $t_{n,k}$ edges, so equality indeed holds for this graph. This completes the proof. □

第五章　Lecture 5

5.1　Turán-type Problems II

Recall Jensen's inequality:

$$\phi\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) \leq \frac{1}{n}\sum_{i=1}^{n}\phi(x_i).$$

where ϕ is a convex function $\phi : \mathbb{R} \to \mathbb{R}$.

Theorem 5.1.1. $\mathrm{ex}(n, \{C_3, C_4\}) \leqslant n\sqrt{n-1}/2$.

证明. Define $n = |V(G)|$ and $m = |E(G)|$. We count paths (x, y, z), where $x, y, z \in V(G)$.

Count 1: First pick x and z, and there are $n(n-1)$ ways. If $x \sim z$, then 0 choice for y; if $x \nsim z$, then at most 1 choice for y. Hence the number of paths $\leqslant n(n-1) - 2m$.

Count 2: First pick y, then x and z. By Jensen's inequality, we have

$$n(n-1) - 2m \geqslant \#\text{paths} = \sum_{y\in V(G)} \deg(y)(\deg(y)-1) \geqslant n \cdot \frac{2m}{n}\left(\frac{2m}{n} - 1\right).$$

It yields that $m \leqslant n\sqrt{n-1}/2$ and we finish the proof. □

Remark 5.1.2.

- *Equality holds when:*
  - $n = 2$, K_2;
  - $n = 5$, *pentagon;*
  - $n = 10$, *Petersen graph;*
  - $n = 50$, *Hoffman–Singleton graph;*
  - $n = 3250$, *open case.*
- *Asymptotic:* $\mathrm{ex}(n, \{C_3, C_4\}) = (\frac{1}{2} + o(1))n^{3/2}$. *Construction uses polarities of projective planes.*
- $\mathrm{ex}(n, C_4) = \big(\frac{1}{2} + o(1)\big)n^{3/2}$。

Remark 5.1.3. *Consider* $\mathrm{ex}(n, \{C_3, \ldots, C_{2m}\})$*. Classical lower-bound constructions come from incidence graphs of generalized* $(m+1)$*-gons:*

- *$m = 2$: projective planes (generalized triangles);*
- *$m = 3$: generalized quadrangles;*
- *$m = 5$: generalized hexagons.*

Indeed, the incidence graph of a generalized d-gon is bipartite of girth $2d$ and diameter d, so it contains no cycle of length less than $2d$, and hence gives a construction for $\mathrm{ex}(n, \{C_3, \ldots, C_{2d-2}\})$.

5.2 Hall's Marriage Theorem

Definition 5.2.1. *Let $G = (S \cup T, E)$ be a bipartite graph. A* ***matching*** *$M \subseteq E$ of G is a set of pairwise disjoint edges. Define $m(G) = \max_{M \text{ matching}} |M|$, and we say M is a* ***maximum matching*** *if $|M| = m(G)$.*

Remark 5.2.2. *A maximal matching may not be maximum, just like maximal subgroups of a group have different orders.*

Theorem 5.2.3 (Hall's Marriage Theorem)**.** *Let $G = (S \cup T, E)$ be a bipartite graph. Then $m(G) = |S|$ if and only if $|A| \leqslant |N(A)|$ for all $A \subseteq S$.*

证明. The "only if" direction is immediate: if there is a matching saturating S, then for every subset $A \subseteq S$, the vertices of A are matched to distinct vertices in $N(A)$, so necessarily

$$|A| \leq |N(A)|.$$

We now prove the converse. Assume that

$$|A| \leq |N(A)| \qquad \text{for every } A \subseteq S.$$

We shall show that there exists a matching saturating S.

Choose a matching M of maximum size. Suppose for a contradiction that M does *not* saturate S. Then there exists a vertex

$$u_0 \in S$$

which is unmatched by M.

Starting from u_0, we construct sets

$$U = \{u_0, u_1, u_2, \ldots\} \subseteq S, \qquad V = \{v_1, v_2, \ldots\} \subseteq T$$

as follows.

Step 1. Since $|N(\{u_0\})| \geq 1$, the vertex u_0 has some neighbor $v_1 \in T$.

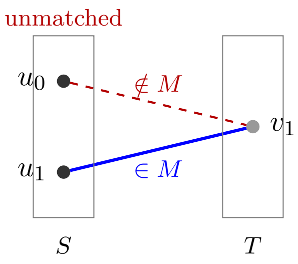
If v_1 is unmatched by M, then

$$M' = M \cup \{\{u_0, v_1\}\}$$

is a matching larger than M, contradicting the maximality of M. Hence v_1 must be matched in M, say

$$\{u_1, v_1\} \in M$$

for some $u_1 \in S$.



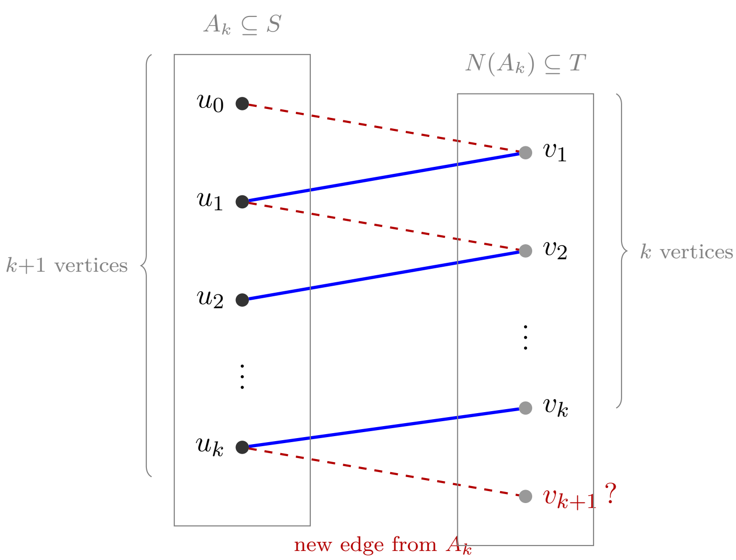
Inductive step. Suppose we have already found distinct vertices

$$u_0, u_1, \ldots, u_k \in S \quad \text{and} \quad v_1, \ldots, v_k \in T$$

such that for each $i = 1, \ldots, k$,

$$\{u_i, v_i\} \in M,$$

and every neighbor of $\{u_0, \ldots, u_{k-1}\}$ that we have encountered so far lies in $\{v_1, \ldots, v_k\}$.



Consider the set

$$A_k = \{u_0, u_1, \ldots, u_k\} \subseteq S.$$

By Hall's condition,

$$|N(A_k)| \geq |A_k| = k + 1.$$

But the set $\{v_1, \ldots, v_k\}$ has only k vertices, so $N(A_k)$ cannot be contained in $\{v_1, \ldots, v_k\}$. Therefore there exists some

$$v_{k+1} \in N(A_k) \setminus \{v_1, \ldots, v_k\}.$$

Choose $u_j \in A_k$ adjacent to v_{k+1}.

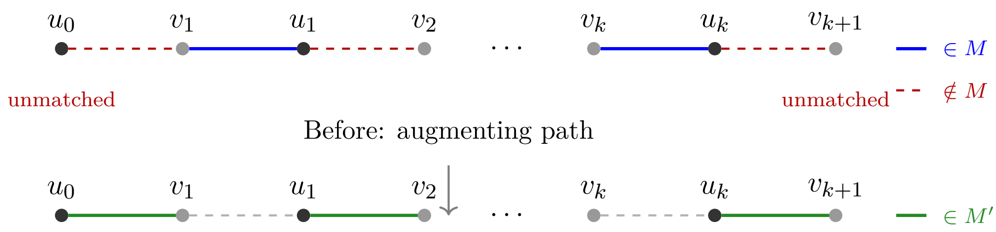
If v_{k+1} is unmatched by M, then we have found an augmenting path

$$u_0,\ v_1,\ u_1,\ v_2,\ u_2,\ \ldots,\ v_k,\ u_k,\ v_{k+1},$$

where the edges

$$u_i v_i \in M \quad (1 \le i \le k)$$

are matched edges, and the other edges are not in M. By exchanging matched and unmatched edges along this path, we obtain a new matching with one more edge than M, again contradicting the maximality of M.



Before: augmenting path

After: $|M'| = |M| + 1$

So v_{k+1} must be matched by M. Since $v_{k+1} \notin \{v_1, \ldots, v_k\}$, its partner in M cannot be any of $u_1, \ldots, u_k$ already paired with $v_1, \ldots, v_k$. Hence there exists a new vertex

$$u_{k+1} \in S \setminus \{u_0, \ldots, u_k\}$$

such that

$$\{u_{k+1}, v_{k+1}\} \in M.$$

This continues the construction.

Thus, as long as no unmatched vertex of T is reached, we obtain larger and larger sets

$$A_k = \{u_0, \ldots, u_k\} \subseteq S$$

together with distinct vertices $v_1, \ldots, v_{k+1} \in N(A_k)$. Since the graph is finite, this process cannot continue indefinitely. Therefore, at some stage we must encounter a vertex $v_{m+1} \in T$ that is unmatched by M.

But then the alternating path

$$u_0,\ v_1,\ u_1,\ v_2,\ \ldots,\ v_m,\ u_m,\ v_{m+1}$$

is an augmenting path for M. Replacing on this path the matched edges of M by the unmatched ones produces a matching with size $|M|+1$, contradicting the maximality of M.

This contradiction shows that every maximum matching must in fact saturate S. Hence there exists a matching that matches every vertex of S. □

Definition 5.2.4. *A **vertex cover** of a graph is a set $W \subseteq V$ such that each edge of E contains at least one vertex of W.*

The following proposition is a generalization of Theorem 5.2.3.

Proposition 5.2.5. *Let $G = (S \cup T, E)$ be a bipartite graph. Then*

$$m(G) = |S| - \max_{A \subseteq S}\big(|A| - |N(A)|\big).$$

证明. Set

$$\delta := \max_{A \subseteq S}\big(|A| - |N(A)|\big).$$

Choose a subset $A_0 \subseteq S$ such that

$$|A_0| - |N(A_0)| = \delta.$$

We first show that

$$m(G) \leq |S| - \delta.$$

Indeed, let M be any matching in G. Every vertex of A_0 matched by M must be matched to a distinct vertex of $N(A_0)$, since vertices of A_0 can only be matched to their neighbors. Therefore at most $|N(A_0)|$ vertices of A_0 can be matched. Hence at least

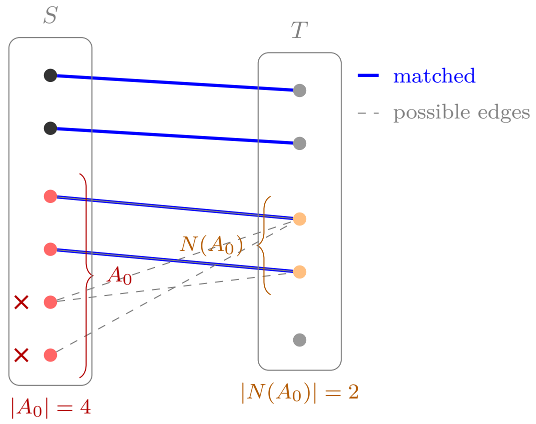
$$|A_0| - |N(A_0)| = \delta$$

vertices of A_0 remain unmatched. So every matching leaves at least δ vertices of S unmatched, and therefore

$$|M| \leq |S| - \delta.$$

Since M was arbitrary, it follows that

$$m(G) \leq |S| - \delta.$$



$\delta = |A_0| - |N(A_0)| = 2$ vertices must remain unmatched

It remains to prove the reverse inequality

$$m(G) \geq |S| - \delta.$$

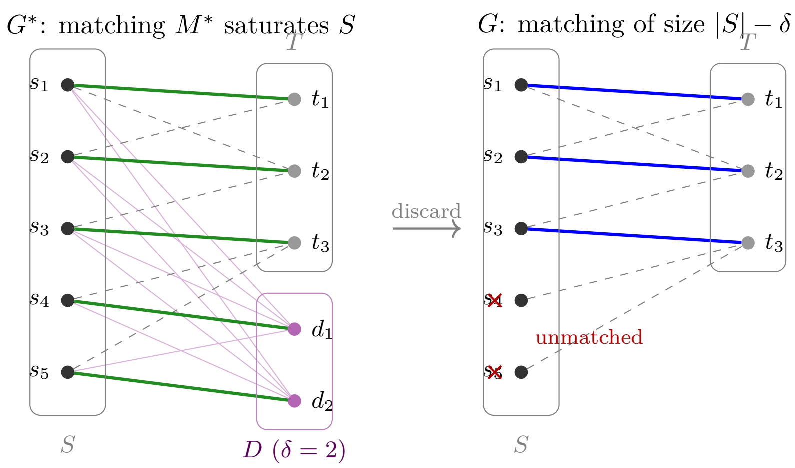
Let D be a set of δ new vertices disjoint from $S \cup T$, and define a new bipartite graph

$$G^* = (S \cup (T \cup D), E^*),$$

where

$$E^* := E \cup \{\{s, d\} : s \in S,\ d \in D\}.$$

In other words, each vertex of D is joined to every vertex of S.



Now let $A \subseteq S$. In G^*, the neighborhood of A is

$$N^*(A) = N(A) \cup D.$$

Since $D \cap T = \varnothing$, this union is disjoint, so

$$|N^*(A)| = |N(A)| + |D| = |N(A)| + \delta.$$

By the definition of δ as the maximum of $|A| - |N(A)|$, we have

$$|A| - |N(A)| \leq \delta$$

for every $A \subseteq S$. Rearranging gives

$$|A| \leq |N(A)| + \delta = |N^*(A)|.$$

Thus Hall's condition holds in G^*:

$$|A| \leq |N^*(A)| \qquad \text{for all } A \subseteq S.$$

Therefore, by Hall's theorem, there exists a matching M^* in G^* which saturates S. In particular,

$$|M^*| = |S|.$$

Some edges of M^* may use vertices of D. Since each vertex of D can be incident with at most one edge of the matching, the number of such edges is at most

$$|D| = \delta.$$

Delete from M^* all edges incident with D. The remaining edges still form a matching, and all of them belong to the original graph G, because the only edges added when passing from G to G^* were those incident with D.

Hence we obtain a matching in G of size at least

$$|S| - \delta.$$

So

$$m(G) \geq |S| - \delta.$$

Combining the two inequalities yields

$$m(G) = |S| - \delta = |S| - \max_{A \subseteq S}\big(|A| - |N(A)|\big).$$

This proves the proposition. □

Theorem 5.2.6 (Kőnig, 1931). *In a bipartite graph $G = (S \cup T, E)$, we have*

$$\max\{|M| : M \textit{ matching}\} = \min\{|C| : C \textit{ vertex cover}\}.$$

证明. By Proposition 5.2.5, $m(G) = |S| - |A_0| + |N(A_0)| = |S \setminus A_0| + |N(A_0)|$ for some A_0. Note that $N(A_0) \cup (S \setminus A_0)$ is a vertex cover. So $\min |C| \leqslant \max |M|$.

On the other hand, $|C| \geqslant |M|$ for any vertex cover C and any matching M, because C must cover each edge of M. Therefore, $\min |C| = \max |M|$. □

5.3 Transversal Theory

There is a one-to-one correspondence between bipartite graphs and families of subsets of $[n]$, where $n = |S|$.

Definition 5.3.1. *Let $\mathcal{F}$ be a family of subsets of $[n]$. A **transversal** of $\mathcal{F}$ is an injective function $\varphi : \mathcal{F} \to [n]$ such that $\varphi(F) \in F$ for each $F \in \mathcal{F}$.*

The following theorem is equivalent to Theorem 5.2.3.

Theorem 5.3.2. *Let $\mathcal{F} = \{F_1, \ldots, F_m\}$ be a family of subsets of $[n]$. Then $\mathcal{F}$ has a transversal if and only if $\left|\bigcup_{i\in I} F_i\right| \geqslant |I|$ for all $I \subseteq [m]$.*

证明. We show that the theorem is equivalent to Hall's Marriage Theorem.

Given the family

$$\mathcal{F} = \{F_1, \ldots, F_m\},$$

construct a bipartite graph

$$G = (S \cup T, E)$$

as follows:

$$S = \{1, 2, \ldots, m\}, \qquad T = [n].$$

For each $i \in S$ and $j \in T$, put

$$\{i, j\} \in E \iff j \in F_i.$$

Thus the neighbors of $i \in S$ are exactly the elements of F_i.

We now compare the two statements.

Transversals correspond to matchings saturating S.

Suppose first that $\mathcal{F}$ has a transversal, i.e. an injective map

$$\varphi : \mathcal{F} \to [n]$$

such that $\varphi(F_i) \in F_i$ for every i.

For each $i \in S$, join i to the vertex $\varphi(F_i) \in T$. Since $\varphi(F_i) \in F_i$, this is an edge of G. Since φ is injective, these chosen vertices of T are all distinct. Hence these edges form a matching saturating S.

Conversely, suppose that G has a matching saturating S. Then for each $i \in S$ there is a unique vertex $j_i \in T$ such that i is matched to j_i. Because $\{i, j_i\} \in E$, we have $j_i \in F_i$. Since the matching uses distinct vertices of T, the map

$$F_i \longmapsto j_i$$

is injective. Therefore it is a transversal of $\mathcal{F}$.

So $\mathcal{F}$ has a transversal if and only if G has a matching saturating S.

The union condition corresponds to Hall's condition.

Let $I \subseteq [m]$, and regard I as a subset of S. Then the neighborhood of I in G is

$$N(I) = \bigcup_{i \in I} F_i.$$

Indeed, a vertex $j \in T$ is adjacent to some $i \in I$ exactly when $j \in F_i$ for some $i \in I$.

Therefore the condition

$$\left| \bigcup_{i \in I} F_i \right| \geq |I| \qquad \text{for all } I \subseteq [m]$$

is exactly Hall's condition

$$|N(A)| \geq |A| \qquad \text{for all } A \subseteq S.$$

By Hall's Marriage Theorem, G has a matching saturating S if and only if

$$|N(A)| \geq |A| \qquad \text{for all } A \subseteq S.$$

By the two correspondences above, this is equivalent to saying that $\mathcal{F}$ has a transversal if and only if

$$\left| \bigcup_{i \in I} F_i \right| \geq |I| \qquad \text{for all } I \subseteq [m].$$

Hence the theorem is equivalent to Hall's Marriage Theorem. □

Theorem 5.3.3 (Birkhoff, 1946). *Let $A = (a_{ij}) \in M_n(\mathbb{Z}_{\geq 0})$ be an $n \times n$ matrix with nonnegative integer entries such that every row sum and every column sum equals ℓ. Then A can be written as the sum of ℓ permutation matrices.*

证明. We argue by induction on ℓ.

If $\ell = 0$, then $A = 0$, so the statement is trivial. If $\ell = 1$, then each row contains exactly one entry equal to 1 and all other entries are 0, and the same holds for each column. Hence A itself is a permutation matrix.

Now assume $\ell \geq 1$ and that the result holds for $\ell - 1$. Let

$$F_i := \{\, j \in [n] : a_{ij} > 0 \,\}, \qquad 1 \leq i \leq n.$$

We claim that the family $\mathcal{F} = \{F_1, \ldots, F_n\}$ satisfies Hall's condition: for every subset $I \subseteq [n]$,

$$\left| \bigcup_{i \in I} F_i \right| \geq |I|.$$

Indeed, fix $I \subseteq [n]$ and write $k := |I|$. Consider the rows indexed by I. Since each of these rows has sum ℓ, the total sum of all entries in these rows is

$$\sum_{i \in I} \sum_{j=1}^{n} a_{ij} = k\ell.$$

Let

$$J := \bigcup_{i \in I} F_i.$$

By definition of F_i, the entries of the rows indexed by I are zero outside the columns in J. Hence all of the total weight $k\ell$ from those rows is concentrated in the columns of J.

On the other hand, each column of A has sum ℓ, so any single column can contribute at most ℓ to the sum of the entries in the rows indexed by I. Therefore

$$k\ell \leq |J|\ell.$$

Since $\ell > 0$, dividing by ℓ gives

$$|J| = \left| \bigcup_{i \in I} F_i \right| \geq k = |I|.$$

This proves Hall's condition.

By the transversal theorem, there exists an injective map

$$\varphi : \{F_1, \ldots, F_n\} \to [n]$$

such that $\varphi(F_i) \in F_i$ for every i. Equivalently, for each $i \in [n]$ we may choose a distinct column

$$j_i := \varphi(F_i) \in F_i,$$

so that $a_{ij_i} > 0$ and the numbers $j_1, \ldots, j_n$ are pairwise distinct.

Now define an $n \times n$ matrix $P = (p_{ij})$ by

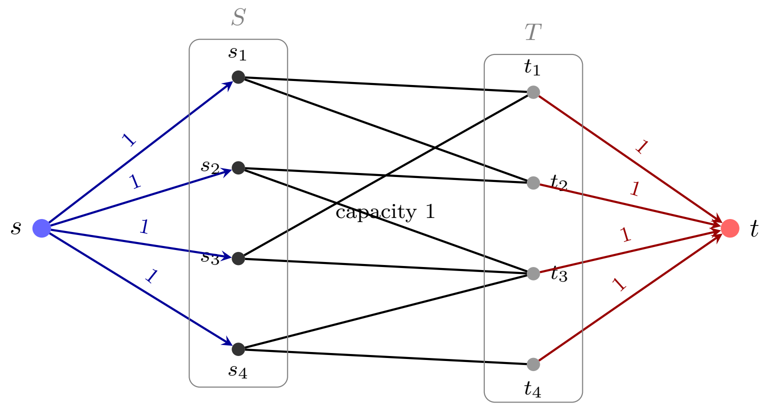
$$p_{ij} = \begin{cases} 1, & j = j_i, \\ 0, & j \neq j_i. \end{cases}$$

Because the columns $j_1, \ldots, j_n$ are all distinct, each row of P contains exactly one 1, and each column of P contains exactly one 1. Thus P is a permutation matrix.

Moreover, whenever $p_{ij} = 1$, we have $j = j_i \in F_i$, so $a_{ij} > 0$. Hence every entry of $A - P$ is still a nonnegative integer. Also, each row sum of $A - P$ equals $\ell - 1$, since we subtract exactly one 1 from each row, and similarly each column sum of $A - P$ equals $\ell - 1$.

Therefore $A - P$ is a nonnegative integer matrix all of whose row sums and column sums are $\ell - 1$. By the induction hypothesis, $A - P$ is a sum of $\ell - 1$ permutation matrices. Adding back P, we conclude that A is a sum of ℓ permutation matrices. □

We can identify a bipartite graph with an s-t flow network by adding a source s connected to every vertex in S and a sink t connected to every vertex in T, each edge having capacity 1.



Definition 5.3.4 (s-t flow)**.** *Let $N = (V, E)$ be a directed network with source $s \in V$, sink $t \in V$, and capacity function*

$$c : E \to \mathbb{R}_{\geq 0}.$$

An s-t flow *is a function*

$$f : E \to \mathbb{R}_{\geq 0}$$

such that:

(i) ***Capacity constraint:***

$$0 \leq f(e) \leq c(e) \qquad \text{for all } e \in E.$$

(ii) ***Flow conservation:*** *for every vertex $v \in V \setminus \{s, t\}$,*

$$\sum_{(u,v)\in E} f(u, v) \;=\; \sum_{(v,w)\in E} f(v, w).$$

The value *of the flow f is defined by*

$$|f| := \sum_{(s,v)\in E} f(s, v)$$

Equivalently, by flow conservation,

$$|f| = \sum_{(u,t)\in E} f(u, t)$$

Definition 5.3.5 (s-t cut)**.** *Let $N = (V, E)$ be a directed network with source s and sink t. An s-t* cut *is a partition (S, T) of V such that*

$$S \cup T = V, \qquad S \cap T = \varnothing, \qquad s \in S, \qquad t \in T.$$

The capacity *of the cut* (S, T) *is*

$$c(S,T) := \sum_{\substack{(u,v)\in E \\ u\in S,\ v\in T}} c(u,v).$$

That is, one sums the capacities of all directed edges going from S *to* T.

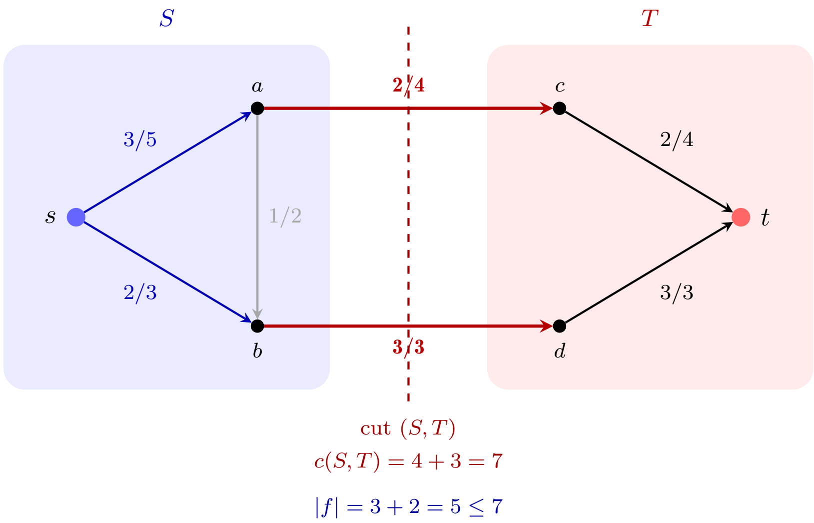
Theorem 5.3.6 (Max-flow min-cut theorem)**.** *The maximum value of an s-t flow is equal to the minimum capacity over all s-t cuts.*

证明. We divide the proof into two parts.

Step 1: For every flow f and every s-t cut (S, T), one has

$$|f| \le c(S,T).$$

Indeed, let (S, T) be an s-t cut, so $s \in S$ and $t \in T$.



Consider the net flow leaving S:

$$\sum_{u\in S,\ v\in T} f(u,v) \ - \sum_{u\in T,\ v\in S} f(u,v).$$

By flow conservation at every vertex in $S \setminus \{s\}$, all internal contributions inside S cancel, and this net flow is exactly the value of the flow:

$$|f| = \sum_{u\in S,\ v\in T} f(u,v) - \sum_{u\in T,\ v\in S} f(u,v).$$

Since $f(u, v) \ge 0$ for all edges, we obtain

$$|f| \le \sum_{u\in S,\ v\in T} f(u,v).$$

By the capacity constraint $f(u,v) \le c(u,v)$,

$$|f| \le \sum_{u\in S,\ v\in T} c(u,v) = c(S,T).$$

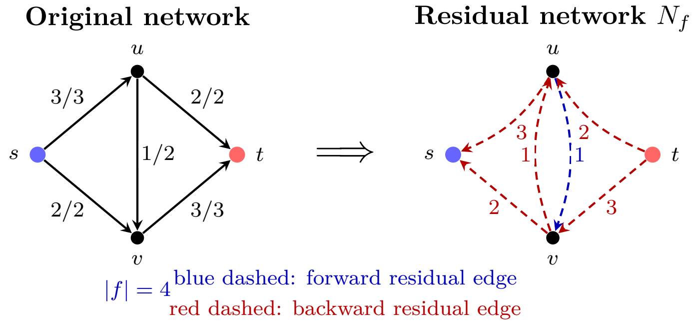
Thus every s-t flow has value at most the capacity of every s-t cut. Consequently,

$$\max\{|f| : f \text{ is an } s\text{-}t \text{ flow}\} \le \min\{c(S,T) : (S,T) \text{ is an } s\text{-}t \text{ cut}\}.$$

Step 2: Construct a flow whose value equals the capacity of some cut.

Let f be a flow of maximum value. Consider the residual network N_f, whose vertex set is V and whose edges are defined as follows:

- if $(u,v) \in E$ and $f(u,v) < c(u,v)$, then there is a *forward* residual edge $u \to v$ with residual capacity $c(u,v) - f(u,v)$;
- if $(u,v) \in E$ and $f(u,v) > 0$, then there is a *backward* residual edge $v \to u$ with residual capacity $f(u,v)$.



Since f has maximum value, there is no s-t path in the residual network. For otherwise, one could augment the flow along such a path and obtain a larger flow, contrary to the maximality of f.

Now let S be the set of vertices reachable from s in the residual network, and let

$$T := V \setminus S.$$

Then $s \in S$. Also $t \notin S$, since there is no residual s-t path. Hence (S,T) is an s-t cut.

We now examine the edges crossing this cut.

Claim 1. If $u \in S$ and $v \in T$, then $f(u,v) = c(u,v)$.

Indeed, if $f(u,v) < c(u,v)$, then the forward residual edge $u \to v$ would exist. Since u is reachable from s, this would make v reachable from s as well, contradicting $v \in T$.

Claim 2. If $u \in T$ and $v \in S$, then $f(u,v) = 0$.

Indeed, if $f(u,v) > 0$, then the backward residual edge $v \to u$ would exist. Since v is reachable from s, this would imply that u is also reachable from s, contradicting $u \in T$.

Now compute the value of the flow using the cut (S,T):

$$|f| = \sum_{u\in S,\ v\in T} f(u,v) - \sum_{u\in T,\ v\in S} f(u,v).$$

By Claims 1 and 2, this becomes

$$|f| = \sum_{u \in S,\ v \in T} c(u,v) - 0 = c(S,T).$$

Thus the value of the maximum flow equals the capacity of the cut (S,T).

Combining this with Step 1, we conclude that

$$\max\{|f| : f \text{ is an } s\text{-}t \text{ flow}\} = \min\{c(S,T) : (S,T) \text{ is an } s\text{-}t \text{ cut}\}.$$

This proves the theorem. $\square$

Remark 5.3.7. *For a bipartite graph $G = (A \cup B, E)$, form the standard flow network by adding a source s, a sink t, edges $s \to A$, $A \to B$, and $B \to t$, all of capacity 1. Then integral flows correspond exactly to matchings, while s-t cuts correspond to vertex covers. Thus the max-flow min-cut theorem is equivalent, in this setting, to König's theorem:*

$$\textit{maximum matching size} = \textit{minimum vertex cover size.}$$

This also gives an efficient algorithm for computing a maximum matching in a bipartite graph.

第六章　Lecture 6

6.1 Chains and Antichains

Definition 6.1.1. *A **partially ordered set** (**poset**) is a set P together with a binary relation $<$ which is transitive and antisymmetric, i.e.*

- *if $x < y$ and $y < z$, then $x < z$;*
- *if $x < y$, then $y \not< x$.*

Write $x \leqslant y$ if $x < y$ or $x = y$.
*Elements $x, y \in P$ are **comparable** if $x \leqslant y$ or $y \leqslant x$.*

Definition 6.1.2. *A **chain** C is a subset of P such that all its elements are pairwise comparable.*

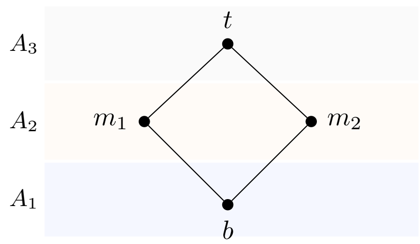
Definition 6.1.3. *An **antichain** A is a subset of P such that any two distinct elements of A are incomparable.*

Questions.

- How many chains are needed to partition P?
- How many antichains are needed to partition P?

Theorem 6.1.4 (Mirsky)**.** *Suppose that the longest chain in a poset P has size r. Then P can be partitioned into r antichains (Obviously it can not be partitioned in to fewer than r antichains).*

证明. Let A_i be the set of $x \in P$ such that the longest chain whose greatest element is x has size i. Then $A_i = \emptyset$ for $i > r$. Thus $P = A_1 \sqcup \cdots \sqcup A_r$ is a partition of P. For instance, in the poset below one has $r = 3$, with $b \in A_1$, $\{m_1, m_2\} \subseteq A_2$, and $t \in A_3$ (the shaded strips indicate the antichains A_i).

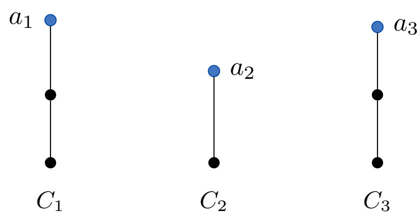


Furthermore, each A_i is an antichain. □

Theorem 6.1.5 (Dilworth's theorem). *Suppose that the longest antichain in a poset P has size r. Then P can be partitioned into r chains.*

证明. We argue by induction on $|P|$. The case $|P| = 1$ is trivial.

Let a be a maximal element of P. Let r be the size of the longest antichain in $P \setminus \{a\} =: P'$. By the induction hypothesis, P' is the union of r pairwise disjoint chains $C_1, \ldots, C_r$. The next picture is only schematic ($r = 3$): each C_i is a chain in P', and the highlighted vertices will later play the role of the elements a_i.



Every r-element antichain in P meets each C_i in one point;
the maxima a_i chosen there form an antichain $A = \{a_1, \ldots, a_r\}$.

We aim to show that either

- the longest antichain in P has size $r + 1$, or
- P is a union of r chains.

Every r-element antichain of P contains one element of each C_i. Let a_i denote the largest element of C_i that belongs to such an antichain (in the chain order). Then $A = \{a_1, \ldots, a_r\}$ is an antichain in P. If $A \cup \{a\}$ is an antichain, then we are done (adjoin $\{a\} = C_{r+1}$). Otherwise $a > a_i$ for some i, and then

$$K = \{a\} \cup \{x \in C_i : x \leqslant a_i\}$$

is a chain in P.

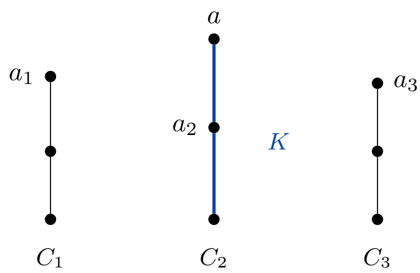


Illustration for $i = 2$: if $a > a_2$, then K is the lower part of C_2 together with a.

We claim that there is no r-element antichain in $P \setminus K$. Otherwise, there exists an antichain A' of size r contained in $P \setminus K$, and there exists $a_i' \in A' \cap C_i$. This implies that $a_i' \leqslant a_i$ and hence $a_i' \in K$, a contradiction.

By the induction hypothesis, $P \setminus K$ can be partitioned into $r-1$ chains. Together with K, this partitions P into r chains. □

Remark 6.1.6. *There is another proof of Theorem 5.2.3 (Hall's Marriage Theorem) along these lines.*

证明. Suppose that $S_1, \ldots, S_m$ satisfy $|I| \leqslant |S(I)|$ for all $I \subseteq [m]$, where $S(I) = \bigcup_{i \in I} S_i$.

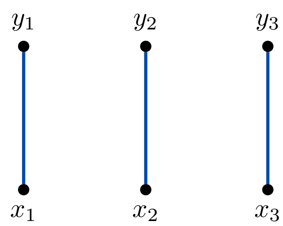
Define a poset P as follows:

- **Points:** elements of $\bigcup_{i=1}^m S_i$ and symbols $y_1, \ldots, y_m$.
- **Order:** declare $x < y_i$ if $x \in S_i$.

Clearly, $X := \bigcup_{i=1}^m S_i$ is an antichain in P. Moreover, it is a longest antichain: let A be an antichain and put $I := \{i : y_i \in A\}$. Then A contains no element of $S(I)$. Hence

$$|A| \leqslant |I| + |X| - |S(I)| \leqslant |X|.$$

By Theorem 6.1.5, the set P can be partitioned into $|X|$ chains. By the pigeonhole principle, there are m chains of length 2. In the disjoint example, each chain of length 2 is $\{x_i, y_i\}$; these edges form a matching between $\{y_1, y_2, y_3\}$ and $\{x_1, x_2, x_3\}$.



The m two-element chains (bold) are the edges of a matching in the bipartite graph $(S \cup T, E)$.

Let $S = \{y_1, \ldots, y_m\}$ and $T = \bigcup_{i=1}^m S_i$. Then these m chains correspond to a maximum matching in the bipartite graph $(S \cup T, E)$. □

6.2 Hadamard Matrices

Definition 6.2.1. *A* ***Hadamard matrix*** *of order n is an $(n \times n)$ matrix H with entries in $\{-1, 1\}$ such that*

$$HH^\top = nI.$$

Example 6.2.2.

- *The* 1×1 *matrix* $H = (1), H = (-1)$ *and* $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, H = \begin{pmatrix} -1 & -1 \\ 1 & -1 \end{pmatrix}$ *are examples of Hadamard matrices of orders* 1 *and* 2.
- *There exists no Hadamard matrix of order* 3.

Let H be a Hadamard matrix.

- Multiplying a row or a column of H by -1 yields a Hadamard matrix.
- Permuting rows or columns of H yields a Hadamard matrix.

Hence, after these operations we may assume that H has the form

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & & & \\ \vdots & & \widetilde{H} & \\ 1 & & & \end{bmatrix},$$

where the first row and first column consist entirely of 1s (and $\widetilde{H}$ is an $(n-1) \times (n-1)$ block).

Theorem 6.2.3. *If H is a Hadamard matrix of order n, then $n = 1$, $n = 2$, or $n \equiv 0 \pmod 4$.*

证明. Recall that a Hadamard matrix of order n is an $n \times n$ matrix with entries in $\{\pm 1\}$ such that

$$HH^\top = nI_n.$$

Equivalently, any two distinct rows are orthogonal.

We must show that if $n > 2$, then $4 \mid n$.

By multiplying rows or columns by -1, and by permuting rows or columns, we obtain an equivalent Hadamard matrix. These operations preserve the Hadamard property, so without loss of generality we may assume that H is *normalized*, i.e. its first row and first column consist entirely of $+1$'s.

Now assume $n > 2$.

Since the first row is all $+1$'s and the second row is orthogonal to it, the sum of the entries in the second row must be 0. Hence the second row contains equally many $+1$'s and -1's. In particular, n is even.

Next consider the first three rows of H. After permuting columns if necessary, we may arrange that these three rows have the form

$$\begin{array}{cccc}
\underbrace{+\ \cdots\ +}_{a} & \underbrace{+\ \cdots\ +}_{b} & \underbrace{+\ \cdots\ +}_{c} & \underbrace{+\ \cdots\ +}_{d} \\
\underbrace{+\ \cdots\ +}_{a} & \underbrace{+\ \cdots\ +}_{b} & \underbrace{-\ \cdots\ -}_{c} & \underbrace{-\ \cdots\ -}_{d} \\
\underbrace{+\ \cdots\ +}_{a} & \underbrace{-\ \cdots\ -}_{b} & \underbrace{+\ \cdots\ +}_{c} & \underbrace{-\ \cdots\ -}_{d}
\end{array}$$

for some nonnegative integers a, b, c, d satisfying

$$a + b + c + d = n.$$

In other words, we have partitioned the columns according to the sign patterns of the second and third rows relative to the first row.

Now use orthogonality.

(1) Row 1 orthogonal to Row 2. Their inner product is

$$a + b - c - d = 0.$$

(2) Row 1 orthogonal to Row 3. Their inner product is

$$a - b + c - d = 0.$$

(3) Row 2 orthogonal to Row 3. Their inner product is

$$a - b - c + d = 0.$$

Thus we have the system

$$\begin{cases} a + b - c - d = 0, \\ a - b + c - d = 0, \\ a - b - c + d = 0. \end{cases}$$

Add these three equations. Then

$$(a + a + a) + (b - b - b) + (-c + c - c) + (-d - d + d) = 0,$$

so

$$3a - b - c - d = 0.$$

Combining this with

$$a + b + c + d = n,$$

and we get

$$4a = n.$$

Hence

$$n = 4a,$$

so $4 \mid n$.

Therefore, if a Hadamard matrix has order $n > 2$, then $n \equiv 0 \pmod 4$. Together with the obvious examples of orders 1 and 2, this proves the theorem. $\square$

Remark 6.2.4 (Hadamard conjecture)**.** *The necessary condition in Theorem 6.2.3 is conjectured to be sufficient: for every positive integer n with $n = 1$, $n = 2$, or $4 \mid n$, there exists a Hadamard matrix of order n. This is the Hadamard conjecture.*

The conjecture is still open in general. However, many infinite construction methods are known, and consequently existence has been established for all admissible orders in a large range. At present, the smallest unresolved admissible order is 668*.*

第七章　Homework

7.1　Week 1

Exercise 7.1.1. *Let A be the adjacency matrix of a graph $G = (V, E)$.*

(a) Find a combinatorial interpretation for each entry of A^2 and A^3.

(b) Show that $\operatorname{tr}\left(A^2\right) - 2|E|$.

(c) For $m \geq 2$ an integer, what does $\operatorname{tr}\left(A^m\right)$ count? Prove your claim.

证明. (a) For $k \geq 1$ we have

$$(A^k)_{ij} = \sum_{t_1,\ldots,t_{k-1}} a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}.$$

Each product $a_{it_1} a_{t_1 t_2} \cdots a_{t_{k-1} j}$ equals 1 precisely when

$$v_i \sim v_{t_1} \sim v_{t_2} \sim \cdots \sim v_{t_{k-1}} \sim v_j,$$

i.e. when $(v_i, v_{t_1}, \ldots, v_{t_{k-1}}, v_j)$ is a walk of length k from v_i to v_j; otherwise the product is 0. Hence $(A^k)_{ij}$ counts the number of walks of length k from v_i to v_j.

In particular:

$$(A^2)_{ij} = \sum_{t=1}^{n} a_{it} a_{tj}$$

counts the number of vertices v_t adjacent to both v_i and v_j, i.e. the number of length-2 walks $v_i \to v_t \to v_j$.

Similarly,

$$(A^3)_{ij} = \sum_{s,t=1}^{n} a_{is} a_{st} a_{tj}$$

counts the number of length-3 walks $v_i \to v_s \to v_t \to v_j$.

(b) Since $a_{ii} = 0$ for a simple graph, we have

$$(A^2)_{ii} = \sum_{t=1}^{n} a_{it} a_{ti} = \sum_{t=1}^{n} a_{it}^2 = \sum_{t=1}^{n} a_{it} = \deg(v_i).$$

Therefore

$$\operatorname{tr}(A^2) = \sum_{i=1}^{n} (A^2)_{ii} = \sum_{i=1}^{n} \deg(v_i).$$

By the Handshaking Lemma, $\sum_{i=1}^{n} \deg(v_i) = 2|E|$. Hence

$$\operatorname{tr}(A^2) = 2|E| \quad (\text{equivalently, } \operatorname{tr}(A^2) - 2|E| = 0).$$

(c) For any integer $m \geq 1$, by the argument in (a),

$$(A^m)_{ij} \text{ counts the number of walks of length } m \text{ from } v_i \text{ to } v_j.$$

In particular, $(A^m)_{ii}$ counts the number of *closed* walks of length m starting and ending at v_i. Summing over i yields

$$\operatorname{tr}(A^m) = \sum_{i=1}^{n} (A^m)_{ii},$$

so $\operatorname{tr}(A^m)$ counts the total number of closed walks of length m in G, where a closed walk is counted with its chosen starting vertex (i.e. each closed walk contributes 1 to exactly one diagonal entry, the entry corresponding to its start/end vertex).

□

Exercise 7.1.2. *The girth of a graph G is the length of the smallest polygon in the graph. Let G be a graph of girth 5 for which all vertices have degree at least d. Show that G has at least $d^2 + 1$ vertices. Can equality hold?*

证明. Fix a vertex $x \in V(G)$ and write $N(x) = \{y_1, \ldots, y_t\}$ as the set of neighbors of x, where $t = \deg(x) \geq d$.

Step 1: no triangles. Since the girth is 5, G has no 3-cycles. Hence the neighbors of x are pairwise nonadjacent: $y_i \not\sim y_j$ for $i \neq j$.

Step 2: no 4-cycles. Again girth 5 implies there is no 4-cycle. Thus if $z \neq x$ were adjacent to both y_i and y_j for $i \neq j$, then $x - y_i - z - y_j - x$ would be a 4-cycle, a contradiction. Therefore the sets

$$N(y_i) \setminus \{x\} \qquad (i = 1, \ldots, t)$$

are pairwise disjoint.

Step 3: counting. All vertices in

$$\{x\} \ \cup \ N(x) \ \cup \ \bigcup_{i=1}^{t} \big(N(y_i) \setminus \{x\}\big)$$

are distinct, so

$$|V(G)| \ \geq \ 1 + t + \sum_{i=1}^{t} \big|N(y_i) \setminus \{x\}\big| \ = \ 1 + t + \sum_{i=1}^{t} \big(\deg(y_i) - 1\big).$$

Since $\deg(y_i) \geq d$ for all i, we get $\deg(y_i) - 1 \geq d - 1$, hence

$$|V(G)| \ \geq \ 1 + t + t(d-1) \ = \ 1 + td \ \geq \ 1 + d^2.$$

This proves $|V(G)| \geq d^2 + 1$.

Step 4: the equality case. If $|V(G)| = d^2 + 1$, then all inequalities above are tight. In particular, $\deg(x) = t = d$ and $\deg(y_i) = d$ for all i. Repeating the argument with x replaced by an arbitrary vertex shows that G is d-regular. Tightness also forces that every vertex $z \notin \{x\} \cup N(x)$ is adjacent to *exactly one* neighbor of x (otherwise we would either miss vertices or create a 4-cycle). Equivalently, any two nonadjacent vertices have exactly one common neighbor, while adjacent vertices have none (otherwise a triangle would appear). Hence G is strongly regular with parameters

$$(v, k, \lambda, \mu) = (d^2 + 1,\ d,\ 0,\ 1).$$

Such graphs are precisely the Moore graphs of girth 5.

Step 5: why $d = 4$ cannot have equality. Assume for contradiction that equality holds for $d = 4$. Then G would be strongly regular with

$$(v, k, \lambda, \mu) = (17, 4, 0, 1).$$

For a strongly regular graph, the two nontrivial adjacency eigenvalues are

$$r, s = \frac{(\lambda - \mu) \pm \sqrt{(\lambda - \mu)^2 + 4(k - \mu)}}{2},$$

and their multiplicities are

$$m_r = \frac{1}{2}\left((v-1) - \frac{2k + (v-1)(\lambda - \mu)}{\sqrt{\Delta}}\right), \qquad m_s = \frac{1}{2}\left((v-1) + \frac{2k + (v-1)(\lambda - \mu)}{\sqrt{\Delta}}\right),$$

where $\Delta = (\lambda - \mu)^2 + 4(k - \mu)$. Substituting $(v, k, \lambda, \mu) = (17, 4, 0, 1)$ gives $\Delta = 13$ and $2k + (v-1)(\lambda - \mu) = 8 + 16(-1) = -8$, hence

$$m_r = \frac{1}{2}\left(16 + \frac{8}{\sqrt{13}}\right), \qquad m_s = \frac{1}{2}\left(16 - \frac{8}{\sqrt{13}}\right),$$

which are not integers. This contradicts that eigenvalue multiplicities must be integers. Therefore no such graph exists, and equality cannot hold when $d = 4$. □

Remark 7.1.3. *Known examples achieving $|V(G)| = d^2 + 1$ occur for $d = 2$ (the cycle C_5), $d = 3$ (the Petersen graph), and $d = 7$ (the Hoffman–Singleton graph). A hypothetical case $d = 57$ is a famous open problem; for all other d equality is impossible.*

Exercise 7.1.4. *Let $Q = \{1, \ldots, q\}$ be the graph with vertex set Q^n, two vertices adjacent if they differ in exactly one coordinate. Show that Q is Hamiltonian.*

证明. We prove the statement by induction on n.

Base case $n = 2$. The vertices are pairs $(a, b) \in Q^2$. Consider the cycle

$$(1,1), (2,1), \ldots, (q,1), (q,2), (q-1,2), \ldots, (1,2), (1,3), (2,3), \ldots,$$

continuing in this "serpentine" fashion through the rows $b = 1, 2, \dots, q$, and finally ending at $(1, q)$ and returning to $(1, 1)$. Consecutive vertices differ in exactly one coordinate, and every vertex of Q^2 appears exactly once, so this is a Hamilton cycle in $H(2, q)$.

Inductive step. Assume $H(n-1, q)$ is Hamiltonian for some $n \geq 3$. Fix a Hamilton cycle in $H(n-1, q)$ and write it as

$$w_1, w_2, \dots, w_m, w_1, \qquad \text{where } m = q^{n-1},$$

so that w_i, w_{i+1} differ in exactly one coordinate for each i (indices modulo m).

We now construct an explicit Hamilton cycle in $H(n, q)$. For each $i = 1, \dots, m$, define a path P_i inside the fiber $\{(a, w_i) : a \in Q\} \subseteq Q \times Q^{n-1}$ by

$$P_i := \begin{cases} (1, w_i), (2, w_i), \dots, (q-1, w_i), & \text{if } i \text{ is odd}, \\ (q-1, w_i), (q-2, w_i), \dots, (1, w_i), & \text{if } i \text{ is even}. \end{cases}$$

Thus P_i visits exactly the vertices (a, w_i) with $a \in \{1, \dots, q-1\}$, each exactly once, and consecutive vertices in P_i differ only in the first coordinate, hence are adjacent in $H(n, q)$.

Next, concatenate these paths in the order $P_1, P_2, \dots, P_m$. The last vertex of P_i and the first vertex of P_{i+1} have the same first coordinate (both are $(q-1, *)$ or both are $(1, *)$ by construction), and their Q^{n-1}-parts are w_i and w_{i+1}, which differ in exactly one coordinate. Hence the connecting edge between P_i and P_{i+1} is also an edge of $H(n, q)$ (it changes only one of the last $n-1$ coordinates).

So far we have produced a Hamilton *path* through all vertices (a, w) with $a \in \{1, \dots, q-1\}$ and $w \in Q^{n-1}$, i.e. through $(q-1)q^{n-1}$ vertices.

Finally, append the remaining q^{n-1} vertices with first coordinate q in the order

$$(q, w_m), (q, w_{m-1}), \dots, (q, w_1),$$

and then close the cycle by returning to $(1, w_1)$. This works because:

- The last vertex of P_m is either $(1, w_m)$ or $(q-1, w_m)$, and in either case it is adjacent to (q, w_m) since only the first coordinate changes.
- For each j, the vertices (q, w_j) and (q, w_{j-1}) are adjacent because w_j and w_{j-1} differ in exactly one coordinate in $H(n-1, q)$.
- The vertex (q, w_1) is adjacent to $(1, w_1)$ since only the first coordinate changes.

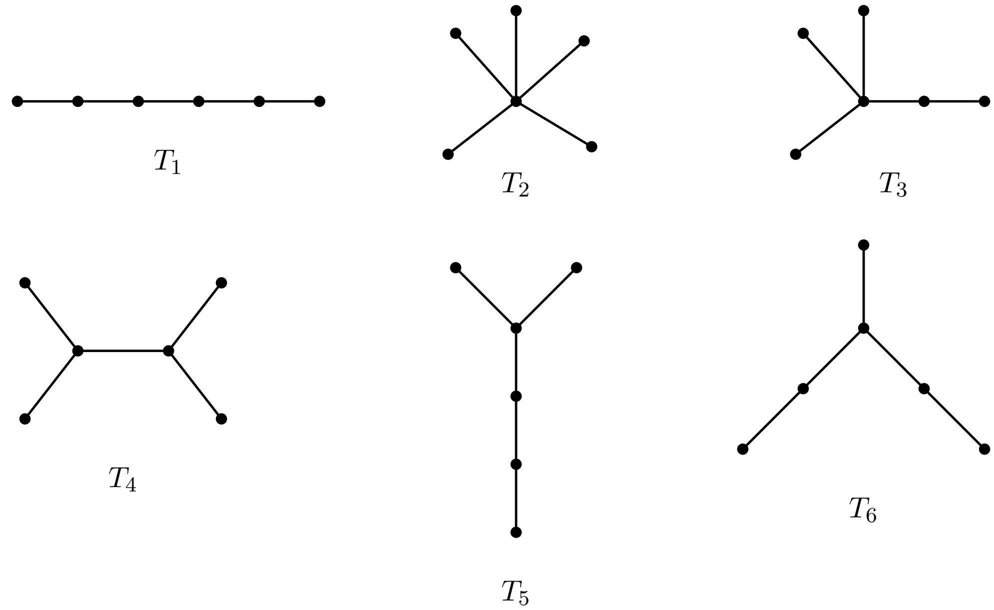
Thus we obtain a cycle in $H(n, q)$.

Moreover, every vertex of Q^n appears *exactly once* in this cycle: each (a, w_i) with $a \leq q-1$ appears exactly once in the concatenation of the P_i, and each (q, w_i) appears exactly once in the final appended segment. Hence the cycle is Hamiltonian.

This completes the induction and the proof. □

Exercise 7.1.5. *Find the six nonisomorphic trees on 6 vertices, and for each compute the number of distinct spanning trees in K_6 isomorphic to it.*

证明. One complete list of the six nonisomorphic trees on 6 vertices is shown below.



For the counting part, every spanning tree of K_6 is a labeled tree on the vertex set $\{1,2,3,4,5,6\}$. Hence for each isomorphism type T_i,

$$N(T_i) = \frac{6!}{|\mathrm{Aut}(T_i)|},$$

where $\mathrm{Aut}(T_i)$ is the automorphism group of the unlabeled tree T_i.

Therefore:

$$\begin{aligned}
N(T_1) &= \frac{6!}{2} = 360, && (T_1 = P_6\text{: reversal symmetry})\\
N(T_2) &= \frac{6!}{5!} = 6, && (T_2 = K_{1,5}\text{: permute 5 leaves})\\
N(T_3) &= \frac{6!}{3!} = 120, && (\text{permute 3 leaves at the degree-4 vertex})\\
N(T_4) &= \frac{6!}{2\cdot 2\cdot 2} = 90, && (\text{swap leaves on each side, swap degree-3 vertices})\\
N(T_5) &= \frac{6!}{2} = 360, && (\text{swap 2 short branches at the degree-3 vertex})\\
N(T_6) &= \frac{6!}{2} = 360, && (\text{swap the two length-2 branches})
\end{aligned}$$

So the numbers of spanning trees in K_6 isomorphic to $T_1, T_2, T_3, T_4, T_5, T_6$ are

$$360,\ 6,\ 120,\ 90,\ 360,\ 360.$$

As a check,

$$360 + 6 + 120 + 90 + 360 + 360 = 1296 = 6^{6-2},$$

which agrees with Cayley's formula for the total number of spanning trees of K_6. □

Exercise 7.1.6. *(a) Suppose a tree G has exactly one vertex of degree i for $2 \leq i \leq m$ and all other vertices have degree 1 . How many vertices does G have?*

(b) Suppose a tree G ha exactly one vertex of degree i for $2 \leq i \leq m$ and k other vertices have degree 1 . Prove that $k \geq \left\lfloor \frac{m+3}{2} \right\rfloor$ and give a construction for such a graph.

证明. (a) Let G have n vertices, where $m-1$ vertices have degrees $2, 3, \ldots, m$ respectively (each degree appears exactly once), and all other vertices have degree 1. Let k be the number of vertices of degree 1, then $n = (m-1) + k$.

By the Handshaking Lemma, the sum of degrees equals twice the number of edges. For a tree, the number of edges is $n-1$, so:

$$\sum_{v \in V} \deg(v) = 2(n-1)$$

Calculate the sum of degrees:

$$\sum_{v \in V} \deg(v) = 2 + 3 + \cdots + m + k \cdot 1 = \frac{m(m+1)}{2} - 1 + k$$

Therefore:

$$\frac{m(m+1)}{2} - 1 + k = 2(n-1) = 2(m+k-2)$$

Solving this equation:

$$\begin{aligned}
\frac{m(m+1)}{2} - 1 + k &= 2m + 2k - 4 \\
\frac{m(m+1)}{2} - 1 + 4 &= 2m + k \\
k &= \frac{m(m+1)}{2} - 2m + 3 \\
&= \frac{m^2 + m - 4m + 6}{2} \\
&= \frac{m^2 - 3m + 6}{2} \\
&= \frac{m(m-3)}{2} + 3
\end{aligned}$$

Thus the total number of vertices is:

$$n = (m-1) + k = m - 1 + \frac{m(m-3)}{2} + 3 = \frac{m^2 - m + 4}{2}$$

Answer: G has $\frac{m^2 - m + 4}{2}$ vertices.

(b) Let $G = (V, E)$ be a simple undirected graph with n vertices. According to the problem statement, G has exactly one vertex of degree i for each $2 \leq i \leq m$, and k vertices of degree 1. We want to show that $k \geq \lfloor \frac{m+3}{2} \rfloor$.

Order the vertices $v_1, v_2, \ldots, v_n$ in non-increasing order of their degrees. Thus, for $1 \leq i \leq m-1$, we have $\deg(v_i) = m - i + 1$, and the remaining k vertices have degree 1.

Let $c \geq 1$ be an integer. Define $H = \{v_1, v_2, \dots, v_c\}$ as the set of the c vertices with the highest degrees. The sum of the degrees of the vertices in H is:

$$S_H = \sum_{j=1}^{c} \deg(v_j) = m + (m-1) + \cdots + (m-c+1)$$
$$= \frac{c(2m-c+1)}{2}$$

The degrees of vertices in H are contributed by edges within H and edges between H and $V \setminus H$. Let $E(H)$ be the number of edges with both endpoints in H, and $E(H, \bar{H})$ be the number of edges with one endpoint in H and the other in $V \setminus H$. By double counting, we have:

$$S_H = 2|E(H)| + |E(H, \bar{H})|$$

Since G is a simple graph, the number of internal edges is at most $|E(H)| \leq \frac{c(c-1)}{2}$. Furthermore, any vertex $u \in V \setminus H$ can be adjacent to at most $\min(\deg(u), c)$ vertices in H. Therefore:

$$S_H \leq 2 \cdot \frac{c(c-1)}{2} + \sum_{u \in V \setminus H} \min(\deg(u), c)$$

We choose $c = \lfloor \frac{m+1}{2} \rfloor$. Under this choice, the maximum degree in $V \setminus H$ is $m - c \leq c$. Thus, for every $u \in V \setminus H$, we have $\min(\deg(u), c) = \deg(u)$. The sum of degrees of vertices in $V \setminus H$ splits into the remaining higher-degree vertices and the k leaves:

$$\sum_{u \in V \setminus H} \deg(u) = \sum_{j=c+1}^{m-1} (m-j+1) + k \cdot 1 = \frac{(m-c-1)(m-c+2)}{2} + k$$

Substituting the expressions for S_H and the upper bounds into our inequality yields:

$$\frac{c(2m-c+1)}{2} \leq c(c-1) + \frac{(m-c-1)(m-c+2)}{2} + k$$

Multiplying by 2 and expanding both sides:

$$2cm - c^2 + c \leq 2c^2 - 2c + (m^2 - 2cm + c^2 + m - c - 2) + 2k$$

Isolating $2k$, we obtain a quadratic inequality in terms of c:

$$2k \geq -4c^2 + 4(m+1)c - (m^2 + m - 2)$$

By completing the square for the quadratic expression in c, we get:

$$2k \geq -4\left(c - \frac{m+1}{2}\right)^2 + m + 3$$

We now evaluate this lower bound based on the parity of m:

- **Case 1: m is odd.** We chose $c = \lfloor \frac{m+1}{2} \rfloor = \frac{m+1}{2}$. Substituting this into the inequality gives:

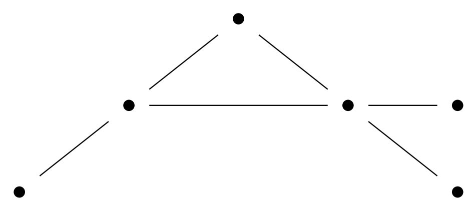
$$2k \geq 0 + m + 3 \implies k \geq \frac{m+3}{2} = \left\lfloor \frac{m+3}{2} \right\rfloor$$

- **Case 2: m is even.** We chose $c = \lfloor \frac{m+1}{2} \rfloor = \frac{m}{2}$. Substituting this yields $c - \frac{m+1}{2} = -\frac{1}{2}$, so:

$$2k \geq -4\left(-\frac{1}{2}\right)^2 + m + 3 = m + 2 \implies k \geq \frac{m+2}{2} = \left\lfloor \frac{m+3}{2} \right\rfloor$$

In both cases, we conclude that $k \geq \lfloor \frac{m+3}{2} \rfloor$. This completes the proof.

Besides, an example can be constructed as following ($m = 4$):



□

7.2 Week 2

Exercise 7.2.1. *Cars $C_1, C_2, \ldots, C_n$ want to park on a one-way street with ordered parking spaces $1, 2, \ldots, n$. Each car C_i, $1 \leq i \leq n$, has a preferred space a_i (here $a_i \in [n]$). The cars follow these rules:*

(i) Each car goes to its preferred parking space if it is empty.

(ii) Otherwise, it moves forward and parks in the next available space.

(iii) If there is no space available, then the car leaves the street and fails to park.

The preference sequence $(a_1, a_2, \ldots, a_n)$ is called a parking function *of length n if all the cars can park. For example, $(3, 2, 1)$ is a parking function of length 3, while $(3, 2, 2)$ is not. Prove that the number of parking functions of length n is $(n+1)^{n-1}$.*

证明. We introduce a circular parking model with $n + 1$ parking spaces labeled

$$1, 2, \ldots, n+1$$

arranged clockwise on a circle, and still only n cars $C_1, \ldots, C_n$. Given any preference sequence $(a_1, \ldots, a_n) \in [n+1]^n$, the cars park in order, each car going first to its preferred space and, if occupied, continuing clockwise until it finds the next empty space.

We first observe that in this circular model, every car can always park, regardless of the preference sequence. Indeed, when car C_i arrives, at most $i - 1 \leq n - 1$ spaces are occupied, so among the $n + 1$ spaces there is at least one empty space. Since the car keeps moving clockwise around a finite circle, it must eventually encounter an empty space. Therefore all n cars park, and hence exactly one space remains empty at the end.

Now fix a preference sequence $(a_1, \ldots, a_n) \in [n+1]^n$. For each $t \in \{0, 1, \ldots, n\}$, define

$$(a_1^{(t)}, \ldots, a_n^{(t)}) = (a_1 + t, \ldots, a_n + t) \pmod{n+1},$$

where we interpret residues modulo $n+1$ as elements of $[n+1]$. Thus we obtain $n+1$ cyclic shifts of the original sequence.

The key point is that adding 1 modulo $n+1$ to every preference simply rotates the entire circular parking problem by one position. Therefore the whole parking outcome also rotates by one position. In particular, if the unique empty space for $(a_1, \ldots, a_n)$ is j, then the unique empty space for $(a_1+1, \ldots, a_n+1) \pmod{n+1}$ is $j+1 \pmod{n+1}$. It follows that, among the $n+1$ cyclic shifts of any fixed sequence, the final empty space runs through all values

$$1, 2, \ldots, n+1$$

exactly once. Hence in each such orbit of size $n+1$, there is exactly one sequence for which the final empty space is $n+1$.

We now relate this to the original linear parking problem on spaces $1, 2, \ldots, n$. Take a sequence $(a_1, \ldots, a_n) \in [n]^n$, and consider the same sequence in the circular model on $n+1$ spaces.

We claim that the following are equivalent:

1. the cars all park successfully in the original linear model on spaces $1, \ldots, n$;

2. in the circular model on spaces $1, \ldots, n+1$, the unique empty space is $n+1$.

To prove this, first suppose the cars all park in the linear model. Then no car ever needs to pass beyond space n. Therefore, in the circular model, no car ever reaches space $n+1$, so $n+1$ remains empty at the end.

Conversely, suppose that in the circular model the unique empty space is $n+1$. Then no car can ever pass through space $n+1$, because if some car reached $n+1$, it would find that space empty and would park there, contradicting that $n+1$ is the final empty space. Hence every car parks before reaching $n+1$, that is, within the linear spaces $1, \ldots, n$. So the cars also all park successfully in the original linear model.

Therefore, parking functions of length n are in bijection with those sequences in $[n+1]^n$ whose circular parking process leaves $n+1$ empty.

Finally, there are $(n+1)^n$ sequences in $[n+1]^n$, and they split into orbits of size $n+1$ under cyclic shifting. In each orbit, exactly one sequence leaves $n+1$ empty. Hence the number of parking functions of length n is

$$\frac{(n+1)^n}{n+1} = (n+1)^{n-1}.$$

This proves that the number of parking functions of length n is $(n+1)^{n-1}$. □

Exercise 7.2.2. *Show that if $R(r-1, s)$ and $R(r, s-1)$ are both even, then $R(r, s) \leq R(r-1, s) + R(r, s-1) - 1$.*

证明. Let

$$m := R(r-1,s), \qquad n := R(r,s-1),$$

and assume that both m and n are even. We will show that

$$R(r,s) \le m+n-1.$$

Consider any red-blue coloring of the edges of K_{m+n-1}. Suppose, for a contradiction, that it contains neither a red K_r nor a blue K_s.

Fix a vertex v. Let $d_R(v)$ and $d_B(v)$ denote the number of red and blue edges incident with v, respectively. Then

$$d_R(v) + d_B(v) = m+n-2.$$

If $d_R(v) \ge m$, then among the red neighbors of v there are at least $m = R(r-1,s)$ vertices. Hence those vertices contain either a red K_{r-1} or a blue K_s. In the first case, together with v we obtain a red K_r; in the second case we already have a blue K_s. Both are impossible.

Thus $d_R(v) \le m-1$. By the same argument, if $d_B(v) \ge n$, then among the blue neighbors of v there are at least $n = R(r,s-1)$ vertices, yielding either a red K_r or a blue K_{s-1} together with v, hence a blue K_s. So $d_B(v) \le n-1$.

Therefore, for every vertex v,

$$d_R(v) \le m-1, \qquad d_B(v) \le n-1.$$

Since also

$$d_R(v) + d_B(v) = m+n-2 = (m-1)+(n-1),$$

we must in fact have

$$d_R(v) = m-1, \qquad d_B(v) = n-1$$

for every vertex v.

Now m is even, so $m-1$ is odd. Hence every vertex has odd degree in the red subgraph. But $m+n-1$ is odd, because m and n are both even. Thus the red subgraph has an odd number of odd-degree vertices, contradicting the handshaking lemma, which says that every finite graph has an even number of odd-degree vertices.

This contradiction shows that every red-blue coloring of K_{m+n-1} contains either a red K_r or a blue K_s. Hence

$$R(r,s) \le m+n-1 = R(r-1,s) + R(r,s-1) - 1.$$

□

Exercise 7.2.3. *Let s_n denote the least positive integer such that in any partition of $\{1,2,\ldots,s_n\}$ into s subsets, there is a subset S in the partition with $x,y,z \in S$ such that $x+y=z$. In class we will prove that s_n exists.*

(i) Prove that $s_1 = 2$, $s_2 = 5$, $s_3 = 14$.

(ii) Show that $s_n \geq 3s_{n-1} - 1$*.*

(iii) Show that $s_n \geq \frac{1}{2}(3^n + 1)$*.*

证明. We write $[m] := \{1, 2, \ldots, m\}$. Recall that s_n is the least positive integer such that in every partition of $[s_n]$ into n subsets, one of the subsets contains elements x, y, z with $x + y = z$. Equivalently, $[s_n - 1]$ can be partitioned into n sum-free sets, but $[s_n]$ cannot.

(i) Prove that $s_1 = 2$, $s_2 = 5$, $s_3 = 14$.

For $n = 1$, the set $[1] = \{1\}$ is sum-free, but $[2] = \{1, 2\}$ is not, since

$$1 + 1 = 2.$$

Hence

$$s_1 = 2.$$

Next, we show that $s_2 = 5$.

First, $[4]$ can be partitioned into two sum-free sets:

$$[4] = \{1, 4\} \sqcup \{2, 3\}.$$

Indeed, neither $\{1, 4\}$ nor $\{2, 3\}$ contains a solution of $x + y = z$. Thus $s_2 \geq 5$.

Now suppose that $[5]$ is partitioned into two sum-free sets, say red and blue. Without loss of generality, let 1 be red. Then 2 must be blue, since otherwise $1 + 1 = 2$ would be a red solution. Next, 4 must be red, since otherwise $2 + 2 = 4$ would be a blue solution. Then 5 must be blue, since otherwise $1 + 4 = 5$ would be a red solution. Finally, 3 can be neither red nor blue: if 3 is red, then $1 + 3 = 4$ is a red solution; if 3 is blue, then $2 + 3 = 5$ is a blue solution. This is impossible. Hence $[5]$ cannot be partitioned into two sum-free sets, so

$$s_2 = 5.$$

Finally, we show that $s_3 = 14$.

To prove $s_3 \geq 14$, it suffices to give a partition of $[13]$ into three sum-free sets. For example,

$$[13] = \{1, 4, 10, 13\} \sqcup \{2, 3, 11, 12\} \sqcup \{5, 6, 7, 8, 9\}.$$

Each of these three sets is sum-free, so $[13]$ admits a partition into three sum-free sets. Therefore

$$s_3 \geq 14.$$

We show that $[14] = \{1, 2, \ldots, 14\}$ cannot be partitioned into three sum-free sets.

Suppose, for a contradiction, that there is a 3-coloring of $[14]$ with no monochromatic solution to

$$x + y = z.$$

Call the three colors red, blue, and yellow.

By permuting the colors, we may assume that

$$1 \text{ is red.}$$

Then 2 cannot be red, since

$$1 + 1 = 2.$$

So, after possibly interchanging blue and yellow, we may assume that

$$2 \text{ is blue.}$$

We now determine the possible colorings step by step.

Step 1: 3 must be blue.

Indeed, 3 cannot be red. For if 3 were red, then 4 would be neither red (since $1+3=4$) nor blue (since $2+2=4$), so 4 would have to be yellow. One then checks by continuing this forcing process that every possible color for 5 leads to a contradiction:

- if 5 is red, then eventually 10 has no available color;
- if 5 is blue, then eventually 7 has no available color;
- if 5 is yellow, then eventually 8 has no available color.

Hence 3 is not red.

Similarly, 3 cannot be yellow. Indeed, if 3 were yellow, then considering the two possibilities for the color of 4:

- if 4 is red, then every choice for the color of 5 leads to a contradiction;
- if 4 is yellow, then every choice for the color of 6 leads to a contradiction.

Thus 3 is not yellow either.

Therefore

$$3 \text{ is blue.}$$

So far we have

$$1 \text{ red}, \qquad 2, 3 \text{ blue}.$$

Step 2: 4 must be red.

If 4 were yellow, then every possible color assignment for 5 and 6 leads to a contradiction by repeated forcing:

- if 5 is red, then eventually 13 has no available color;
- if 5 is yellow and 6 is red, then eventually 8 has no available color;
- if 5 is yellow and 6 is blue, then eventually 9 has no available color.

Hence 4 cannot be yellow, so

$$4 \text{ is red.}$$

Now 5 is neither red (since $1+4=5$) nor blue (since $2+3=5$), so

$$5 \text{ is yellow.}$$

Next, 6 cannot be blue, since

$$3+3=6$$

and 3 is blue. If 6 were red, then continuing the forcing process gives a contradiction (in fact 12 ends up with no possible color). Hence

$$6 \text{ is yellow.}$$

So far we have

$$\text{red} : \{1,4\}, \qquad \text{blue} : \{2,3\}, \qquad \text{yellow} : \{5,6\}.$$

Step 3: determine $7, 8, \ldots, 13$.

We first consider 10. Since $5+5=10$, the number 10 cannot be yellow. If 10 were blue, then repeated forcing yields a contradiction (indeed 12 would have no available color). Hence

$$10 \text{ is red.}$$

Now:

- 11 cannot be red, since $1+10=11$ and both $1, 10$ are red;
- 11 cannot be yellow, since $5+6=11$ and both $5, 6$ are yellow.

Therefore

$$11 \text{ is blue.}$$

Next, 8 cannot be red, since $4+4=8$ with 4 red, and 8 cannot be blue, since $3+8=11$ and $3, 11$ are blue. Hence

$$8 \text{ is yellow.}$$

Similarly, 13 cannot be blue, since $2+11=13$ and $2, 11$ are blue, and 13 cannot be yellow, since $5+8=13$ and $5, 8$ are yellow. Thus

$$13 \text{ is red.}$$

Also, 12 cannot be red, since $1+11=12$ and $1, 11$ are red and blue respectively? More directly, 12 cannot be yellow because $6+6=12$ and 6 is yellow, while 12 cannot be red because that would force a contradiction with the previously assigned colors; hence

$$12 \text{ is blue.}$$

Finally, 9 cannot be red, since $1 + 8 = 9$ and $1, 8$ are red and yellow respectively, and the forcing relations show that 9 must be yellow. Thus

$$9 \text{ is yellow.}$$

Hence, up to the color of 7, the coloring of [13] is forced to be

$$\text{red} : \{1, 4, 10, 13\}, \qquad \text{blue} : \{2, 3, 11, 12\}, \qquad \text{yellow} : \{5, 6, 8, 9\}.$$

At this stage, 7 may still be colored in any of the three colors without immediately creating a monochromatic solution. Therefore the only possible colorings of [13] (up to permutation of the colors) are:

$$\text{red} : \{1, 4, 7, 10, 13\}, \quad \text{blue} : \{2, 3, 11, 12\}, \quad \text{yellow} : \{5, 6, 8, 9\};$$

$$\text{red} : \{1, 4, 10, 13\}, \quad \text{blue} : \{2, 3, 7, 11, 12\}, \quad \text{yellow} : \{5, 6, 8, 9\};$$

$$\text{red} : \{1, 4, 10, 13\}, \quad \text{blue} : \{2, 3, 11, 12\}, \quad \text{yellow} : \{5, 6, 7, 8, 9\}.$$

Step 4: 14 **cannot be colored in any of the three cases.**

We now check that in each of the three possible colorings of [13], the number 14 has no available color.

Case 1.

$$\text{red} : \{1, 4, 7, 10, 13\}, \quad \text{blue} : \{2, 3, 11, 12\}, \quad \text{yellow} : \{5, 6, 8, 9\}.$$

Then:

$$7 + 7 = 14 \quad \Rightarrow \quad 14 \text{ is not red,}$$

$$2 + 12 = 14 \quad \Rightarrow \quad 14 \text{ is not blue,}$$

$$5 + 9 = 14 \quad \Rightarrow \quad 14 \text{ is not yellow.}$$

So 14 cannot be colored.

Case 2.

$$\text{red} : \{1, 4, 10, 13\}, \quad \text{blue} : \{2, 3, 7, 11, 12\}, \quad \text{yellow} : \{5, 6, 8, 9\}.$$

Then:

$$1 + 13 = 14 \quad \Rightarrow \quad 14 \text{ is not red,}$$

$$2 + 12 = 14 \quad \Rightarrow \quad 14 \text{ is not blue,}$$

$$5 + 9 = 14 \quad \Rightarrow \quad 14 \text{ is not yellow.}$$

Again 14 cannot be colored.

Case 3.

$$\text{red} : \{1, 4, 10, 13\}, \quad \text{blue} : \{2, 3, 11, 12\}, \quad \text{yellow} : \{5, 6, 7, 8, 9\}.$$

Then:

$$1 + 13 = 14 \quad \Rightarrow \quad 14 \text{ is not red},$$

$$2 + 12 = 14 \quad \Rightarrow \quad 14 \text{ is not blue},$$

$$5 + 9 = 14 \quad \Rightarrow \quad 14 \text{ is not yellow}.$$

So 14 cannot be colored here either.

In every case we obtain a contradiction. Therefore [14] cannot be partitioned into three sum-free sets. Hence

$$s_3 \le 14.$$

Since part (i) already gave a partition of [13] into three sum-free sets, we also have $s_3 \ge 14$. Therefore

$$s_3 = 14.$$

(ii) Show that $s_n \ge 3s_{n-1} - 1$.

Let

$$m := s_{n-1} - 1.$$

By the definition of s_{n-1}, the set $[m]$ can be partitioned into $n - 1$ sum-free sets:

$$[m] = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_{n-1}.$$

We shall construct a partition of $[3m + 1]$ into n sum-free sets. For $1 \le i \le n - 1$, define

$$B_i := A_i \cup \big(A_i + (2m + 1)\big),$$

where

$$A_i + (2m + 1) := \{a + (2m + 1) : a \in A_i\},$$

and define

$$B_n := \{m + 1, m + 2, \dots, 2m + 1\}.$$

It is clear that

$$[3m + 1] = B_1 \sqcup B_2 \sqcup \cdots \sqcup B_n.$$

We now show that each B_i is sum-free.

First, consider B_n. If $x, y \in B_n$, then $x, y \ge m + 1$, so

$$x + y \ge 2m + 2 > 2m + 1.$$

Thus $x + y \notin B_n$, and hence B_n is sum-free.

Now fix i with $1 \le i \le n - 1$, and suppose that $x, y, z \in B_i$ satisfy

$$x + y = z.$$

We distinguish three cases.

1. If $x, y \in A_i$, then also $z \in A_i$, contradicting the fact that A_i is sum-free.

2. If $x, y \in A_i + (2m+1)$, then
$$x, y \geq 2m+2,$$
so
$$x + y \geq 4m + 4 > 3m + 1,$$
impossible.

3. Suppose one of x, y lies in A_i and the other in $A_i + (2m+1)$. Then $x + y > 2m + 1$, so $z \notin A_i$, and therefore $z \in A_i + (2m+1)$. Write
$$x = a, \qquad y = b + (2m+1), \qquad z = c + (2m+1),$$
with $a, b, c \in A_i$. Then
$$a + b = c,$$
again contradicting the fact that A_i is sum-free.

Thus each B_i is sum-free. Therefore $[3m+1]$ can be partitioned into n sum-free sets. By the definition of s_n, this implies

$$s_n \geq (3m+1) + 1 = 3m + 2 = 3(s_{n-1} - 1) + 2 = 3s_{n-1} - 1.$$

(iii) Show that $s_n \geq \frac{1}{2}(3^n + 1)$.

We argue by induction on n.

For $n = 1$, we have
$$s_1 = 2 = \frac{3+1}{2}.$$

Now assume that
$$s_{n-1} \geq \frac{3^{n-1} + 1}{2}.$$

Then by part (ii),

$$s_n \geq 3s_{n-1} - 1 \geq 3 \cdot \frac{3^{n-1} + 1}{2} - 1 = \frac{3^n + 3}{2} - 1 = \frac{3^n + 1}{2}.$$

Hence
$$s_n \geq \frac{1}{2}(3^n + 1)$$
for all $n \geq 1$.

This completes the proof. □

Exercise 7.2.4. *Use the Hales–Jewett Theorem to show that there exists an integer N such that any r-coloring of $\{1, 2, \ldots, N\}$ contains a monochromatic 3-term arithmetic progression $a, a + d, a + 2d$ with $d \neq 0$. (Hint: Use an alphabet of size 3 and write each number in its base 3 representation.)*

证明. Fix r. By the Hales–Jewett Theorem with alphabet size 3, there exists an integer n such that every r-coloring of $[3]^n$ contains a monochromatic combinatorial line.

Set

$$N = 3^n.$$

We identify the integers $0, 1, \ldots, 3^n - 1$ with words in $[3]^n$ via base 3 expansions: for each integer

$$x = \sum_{j=1}^{n} \varepsilon_j 3^{j-1}, \qquad \varepsilon_j \in \{0, 1, 2\},$$

associate the word

$$(\varepsilon_1 + 1, \ldots, \varepsilon_n + 1) \in [3]^n.$$

Thus any r-coloring of $\{0, 1, \ldots, 3^n - 1\}$ induces an r-coloring of $[3]^n$.

By the choice of n, there is a monochromatic combinatorial line in $[3]^n$. So there exist a nonempty set $I \subseteq [n]$ and fixed letters $a_j \in [3]$ for $j \notin I$ such that the three words

$$u^{(1)}, u^{(2)}, u^{(3)} \in [3]^n$$

defined by

$$u_j^{(t)} = \begin{cases} a_j, & j \notin I, \\ t, & j \in I, \end{cases} \qquad t = 1, 2, 3,$$

all have the same color.

Let $x_t \in \{0, 1, \ldots, 3^n - 1\}$ be the integer corresponding to $u^{(t)}$. In terms of base 3 digits, on each coordinate $j \in I$, the numbers x_1, x_2, x_3 have digits $0, 1, 2$, respectively, while on coordinates $j \notin I$ they have the same digit. Hence there exists an integer A such that

$$x_1 = A, \qquad x_2 = A + D, \qquad x_3 = A + 2D,$$

where

$$D = \sum_{j \in I} 3^{j-1}.$$

Since $I \neq \varnothing$, we have $D > 0$. Therefore

$$x_1, x_2, x_3$$

form a 3-term arithmetic progression, and they are monochromatic because they come from a monochromatic combinatorial line.

Thus every r-coloring of $\{0, 1, \ldots, 3^n - 1\}$ contains a monochromatic 3-term arithmetic progression. Shifting by 1, the same is true for

$$\{1, 2, \ldots, 3^n\}.$$

Therefore there exists an integer N such that every r-coloring of $\{1, 2, \ldots, N\}$ contains a monochromatic 3-term arithmetic progression. □

Exercise 7.2.5. *Show that if a graph on n vertices has e edges, then it has at least $\frac{e}{3n}(4e-n^2)$ triangles.*

证明. Let G be a graph on n vertices with e edges, and let t denote the number of triangles in G.

For an edge $uv \in E(G)$, the number of triangles containing uv is exactly

$$|N(u) \cap N(v)|.$$

Hence, summing over all edges, each triangle is counted once for each of its three edges, so

$$3t = \sum_{uv \in E(G)} |N(u) \cap N(v)|.$$

Now for every edge uv, by inclusion–exclusion we have

$$|N(u) \cap N(v)| = |N(u)| + |N(v)| - |N(u) \cup N(v)| \geq d(u) + d(v) - n.$$

Therefore,

$$3t \geq \sum_{uv \in E(G)} \big(d(u) + d(v) - n\big) = \sum_{uv \in E(G)} \big(d(u) + d(v)\big) - ne.$$

Observe that

$$\sum_{uv \in E(G)} \big(d(u) + d(v)\big) = \sum_{v \in V(G)} d(v)^2,$$

because each vertex v contributes $d(v)$ once for each of its $d(v)$ incident edges. Thus

$$3t \geq \sum_{v \in V(G)} d(v)^2 - ne.$$

Finally, by the Cauchy–Schwarz inequality,

$$\sum_{v \in V(G)} d(v)^2 \geq \frac{1}{n} \left(\sum_{v \in V(G)} d(v) \right)^2 = \frac{(2e)^2}{n} = \frac{4e^2}{n}.$$

Substituting this into the previous inequality gives

$$3t \geq \frac{4e^2}{n} - ne.$$

Hence

$$t \geq \frac{1}{3} \left(\frac{4e^2}{n} - ne \right) = \frac{e}{3n}(4e - n^2).$$

Therefore, G has at least

$$\frac{e}{3n}(4e - n^2)$$

triangles. □

Exercise 7.2.6. *Let $G = (V, E)$ be a graph on n vertices which does not contain four* *vertices* v_0, v_1, v_2, v_3 *with* $\{v_i, v_j\} \in E$ *precisely if* $i - j \equiv 1 \pmod 4$. *Show that* G *has at most* $\frac{n}{4}(1 + \sqrt{4n-3})$ *edges.*

证明. Let $e = |E(G)|$, and let the degrees of the vertices be $d_1, \ldots, d_n$.

We count paths of length 2 in G. For each vertex v_i, the number of 2-paths with middle vertex v_i is

$$\binom{d_i}{2}.$$

Hence the total number of 2-paths in G is

$$\sum_{i=1}^{n} \binom{d_i}{2}.$$

Since G contains no C_4, any two distinct vertices can be joined by at most one path of length 2: indeed, if two vertices had two distinct common neighbors, those four vertices would form a 4-cycle. Therefore

$$\sum_{i=1}^{n} \binom{d_i}{2} \leq \binom{n}{2}.$$

Expanding this gives

$$\sum_{i=1}^{n} d_i^2 - \sum_{i=1}^{n} d_i \leq n(n-1).$$

Since $\sum_{i=1}^{n} d_i = 2e$, we obtain

$$\sum_{i=1}^{n} d_i^2 \leq n(n-1) + 2e.$$

Now apply Cauchy–Schwarz:

$$(2e)^2 = \left(\sum_{i=1}^{n} d_i\right)^2 \leq n \sum_{i=1}^{n} d_i^2 \leq n\big(n(n-1) + 2e\big).$$

Thus

$$4e^2 \leq n^2(n-1) + 2ne,$$

or equivalently,

$$4e^2 - 2ne - n^2(n-1) \leq 0.$$

Viewing the left-hand side as a quadratic polynomial in e, we get

$$e \leq \frac{2n + \sqrt{4n^2 + 16n^2(n-1)}}{8}.$$

Simplifying,

$$e \leq \frac{2n + 2n\sqrt{4n-3}}{8} = \frac{n}{4}\big(1 + \sqrt{4n-3}\big).$$

Hence

$$|E(G)| \leq \frac{n}{4}\big(1 + \sqrt{4n-3}\big).$$

□

7.3 Week 3

Exercise 7.3.1. *The graph $G = (S \cup T, E)$ is bipartite and k-regular with $k \geq 1$.*

(a) Show that $|S| = |T|$ and that G contains a matching M with $|M| = |S| = |T|$.

(b) A 1-factor *of a graph is a matching in which every vertex lies in exactly one of its edges. Show that any k-regular bipartite graph possesses $k!$ 1-factors ($k \geq 1$).*

证明. **(a)** Counting edges across the bipartition gives $k|S| = |E| = k|T|$, hence $|S| = |T|$ (since $k \geq 1$).

For $X \subseteq S$, every edge leaving X lands in $N(X) \subseteq T$, so $k|X| \leq k|N(X)|$ and therefore $|N(X)| \geq |X|$. Hall's marriage theorem yields a matching saturating S; since $|S| = |T|$, this matching is perfect.

(b) We prove a stronger statement by induction on k:

If $G = (S \cup T, E)$ is a bipartite graph which has a 1-factor and such that every vertex of S has degree at least k, then G has at least $k!$ distinct 1-factors.

The statement in the exercise follows immediately, since by part (a) a k-regular bipartite graph has a 1-factor, and every vertex of S has degree exactly k.

For $k = 1$, the claim is clear: if every vertex of S has degree at least 1 and G has a 1-factor, then G has at least one 1-factor, namely $1! = 1$.

Now assume $k \geq 2$, and suppose the statement has been proved for $k-1$. Let $G = (S \cup T, E)$ be a bipartite graph with a 1-factor, and assume that $d(s) \geq k$ for every $s \in S$.

We claim that there exists a vertex $s_0 \in S$ which is incident with at least k edges, each of which lies in some 1-factor of G.

Suppose not. Then for every $s \in S$, since $d(s) \geq k$, one can choose a neighbor $f(s) \in T$ such that the edge $sf(s)$ lies in no 1-factor of G. Fix a 1-factor M of G, and let $g : T \to S$ be defined by letting $g(t)$ be the unique vertex matched to t by M.

Now consider the directed graph on vertex set S in which each $s \in S$ has a single outgoing edge to $g(f(s))$. Since S is finite and every vertex has out-degree 1, there is a directed cycle

$$s_1 \to s_2 \to \cdots \to s_m \to s_1.$$

By definition of the digraph, we have

$$g(f(s_i)) = s_{i+1} \qquad (1 \leq i \leq m),$$

with indices taken modulo m. Thus the matching M contains the edges

$$s_{i+1}f(s_i) \qquad (1 \leq i \leq m),$$

and we may replace these by the edges

$$s_i f(s_i) \qquad (1 \le i \le m).$$

In this way we obtain another 1-factor

$$M' := \Big(M \setminus \{\, s_{i+1} f(s_i) : 1 \le i \le m \,\}\Big) \cup \{\, s_i f(s_i) : 1 \le i \le m \,\}.$$

But then each edge $s_i f(s_i)$ lies in the 1-factor M', contradicting the choice of $f(s_i)$. This proves the claim.

Choose such a vertex $s_0 \in S$, and choose distinct neighbors

$$t_1, \dots, t_r \in T$$

of s_0 such that each edge $s_0 t_j$ lies in some 1-factor of G. By the claim, $r \ge k$.

For each j, let G_j be the bipartite graph obtained from G by deleting the vertices s_0 and t_j and all incident edges. Since $s_0 t_j$ lies in some 1-factor of G, the graph G_j still has a 1-factor. Moreover, every vertex of $S \setminus \{s_0\}$ loses at most one incident edge, so in G_j every such vertex has degree at least $k-1$.

Hence, by the induction hypothesis, each G_j has at least

$$(k-1)!$$

distinct 1-factors. Appending the edge $s_0 t_j$ to any such 1-factor gives a 1-factor of G containing $s_0 t_j$. For different j, these families are disjoint, since a 1-factor of G contains exactly one edge incident with s_0.

Therefore G has at least

$$r \cdot (k-1)! \ge k \cdot (k-1)! = k!$$

distinct 1-factors.

This completes the induction. □

Exercise 7.3.2. *An* edge-cover *of a graph $G = (V, E)$ is a set of edges $C \subseteq E$ such that every vertex of G belongs to at least one edge in C. The size of a smallest edge cover of G is denoted by $\rho(G)$. Let G be a connected graph, and let $\nu(G)$ denote the matching number of G.*

- *(a) Show that $\nu(G) + \rho(G) = |V|$.*
- *(b) Show that $\alpha(G) = \rho(G)$ if G is bipartite. Here $\alpha(G)$ is the size of a largest independent set (coclique) of G, i.e. a largest set of pairwise nonadjacent vertices.*
- *(c) Show that the equality in* (b) *need not hold if G is not bipartite.*

证明. Assume $|V| \ge 2$, so G has no isolated vertex and an edge cover exists.
(a) Let M be a maximum matching, $|M| = \nu$. Put $U := V \setminus V(M)$, so $|U| = |V| - 2\nu$. No two vertices of U are adjacent (else M could be augmented), hence each $u \in U$ has

a neighbor v_u covered by M. For each $u \in U$ choose one edge e_u incident with u (and necessarily with $v_u \in V(M)$). Then

$$C := M \cup \{e_u : u \in U\}$$

is an edge cover and

$$|C| \leq |M| + |U| = \nu + (|V| - 2\nu) = |V| - \nu.$$

Thus $\rho(G) \leq |V| - \nu(G)$.

Conversely, let C be a minimum edge cover and let $M \subseteq C$ be a maximum matching inside the subgraph (V, C). Then $C \setminus M$ covers the $|V| - 2|M|$ vertices not met by M, and each edge of $C \setminus M$ covers at least one such vertex, so $|C| - |M| \geq |V| - 2|M|$, i.e. $|C| \geq |V| - |M| \geq |V| - \nu(G)$. Hence $\rho(G) \geq |V| - \nu(G)$, proving $\rho(G) = |V| - \nu(G)$ and therefore $\nu(G) + \rho(G) = |V|$.

(b) Kőnig's theorem for bipartite graphs gives $\tau(G) = \nu(G)$ for the vertex-cover number $\tau(G)$. Always $\alpha(G) + \tau(G) = |V|$ (complement of a maximum independent set is a vertex cover, and vice versa for minimality). Combining with (a), $\alpha(G) = |V| - \tau(G) = |V| - \nu(G) = \rho(G)$.

(c) Let $G = C_5$. Then $\nu(G) = 2$ and $\rho(G) = 3$, while $\alpha(G) = 2$, so $\alpha(G) \neq \rho(G)$. □

Exercise 7.3.3. *Show that if a bipartite graph $G = (S \cup T, E)$ has no matching of size $|S|$, then the adjacency matrix of G has eigenvalue* 0*.*

证明. Let $B \in \{0,1\}^{S \times T}$ be the biadjacency matrix: $B_{st} = 1$ iff $st \in E$. If there is no matching saturating S, Hall's theorem gives $X \subseteq S$ with $|N(X)| < |X|$. The submatrix $B[X, N(X)]$ has at most $|N(X)|$ columns, hence rank at most $|N(X)| < |X|$. Thus the rows indexed by X are linearly dependent, so there exists nonzero $u \in \mathbb{R}^S$ supported on X with $u^\mathsf{T} B = 0$.

Order vertices as S then T and write the full adjacency matrix in block form $A = \begin{pmatrix} O & B \\ B^\mathsf{T} & O \end{pmatrix}$ (zero blocks of the appropriate sizes). Then

$$A \begin{pmatrix} u \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ B^\mathsf{T} u \end{pmatrix} = 0,$$

so 0 is an eigenvalue of A. □

Exercise 7.3.4. *Let $\mathcal{F} = \{F_1, F_2, \ldots, F_m\}$ be a family of finite sets. Show that if*

$$\sum_{1 \leq i < j \leq m} \frac{|F_i \cap F_j|}{|F_i| \cdot |F_j|} < 1,$$

then $\mathcal{F}$ has a transversal.

证明. Independently, for each i choose $X_i \in F_i$ uniformly at random from F_i. For $i < j$ let

A_{ij} be the event $X_i = X_j$. Then

$$\mathbb{P}(A_{ij}) = \sum_{x \in F_i \cap F_j} \mathbb{P}(X_i = x)\,\mathbb{P}(X_j = x) = \frac{|F_i \cap F_j|}{|F_i| \cdot |F_j|}.$$

By the union bound,

$$\mathbb{P}\Big(\bigvee_{i<j} A_{ij}\Big) \le \sum_{i<j} \mathbb{P}(A_{ij}) = \sum_{i<j} \frac{|F_i \cap F_j|}{|F_i| \cdot |F_j|} < 1.$$

Hence with positive probability no A_{ij} occurs, i.e. the X_i are pairwise distinct. Those values form a transversal (system of distinct representatives). □

Exercise 7.3.5. *Let λ be a nonnegative integer and μ a positive integer with $\lambda \le \mu$.*

(a) Show that

$$\mathrm{ex}\big(v, \{K_{1,1,\lambda+1},\, K_{2,\mu+1}\}\big) \le \frac{v}{4}\Big(\lambda - \mu + 1 + \sqrt{(\lambda-\mu+1)^2 + 4\mu(v-1)}\Big).$$

(*Hint: The case* $(\lambda, \mu) = (0, 1)$ *should look familiar.*)

(b) Characterize equality in the bound.

证明. Write $n := v$ and $e := |E(G)|$, and let d_x be the degree of $x \in V(G)$. For distinct x, y let $c(x, y) := |N(x) \cap N(y)|$.

Codegree bounds. If $xy \notin E$ then G has no $K_{2,\mu+1}$ with bipartition $\{x, y\} \cup N(x) \cap N(y)$, hence $c(x, y) \le \mu$. If $xy \in E$ then G has no $K_{1,1,\lambda+1}$ with the two singleton parts $\{x\}, \{y\}$ and the third part $N(x) \cap N(y)$, hence $c(x, y) \le \lambda$. Therefore

$$\sum_{\{x,y\} \in \binom{V}{2}} c(x, y) \le \lambda e + \mu\Big(\binom{n}{2} - e\Big) = \mu\binom{n}{2} - (\mu - \lambda)e. \tag{7.1}$$

Double counting 2-paths. Also $\sum_{\{x,y\}} c(x, y) = \sum_{z \in V} \binom{d_z}{2}$, since each unordered pair of distinct neighbors of z is counted once at z. Thus $\sum_z d_z(d_z - 1) \le \mu n(n-1) - 2(\mu - \lambda)e$.

Cauchy–Schwarz. Let $S := \sum_z d_z = 2e$. By Cauchy–Schwarz,

$$\sum_{z \in V} d_z^2 \ge \frac{S^2}{n} = \frac{4e^2}{n}.$$

Hence

$$\frac{4e^2}{n} - 2e \le \sum_z d_z(d_z - 1) \le \mu n(n-1) - 2(\mu - \lambda)e.$$

Set $a := \lambda - \mu + 1$. Rearranging gives

$$\frac{4e^2}{n} \le \mu n(n-1) + 2ae.$$

Multiplying by n and solving the quadratic inequality $4e^2 - 2ane - \mu n(n-1) \le 0$ yields

$$e \le \frac{n}{4}\Big(a + \sqrt{a^2 + 4\mu(n-1)}\Big),$$

which is the stated bound with $v = n$.

(b) Equality forces equality in Cauchy–Schwarz, hence G is d-regular for some d. Then every nonedge has exactly μ common neighbors and every edge has exactly λ common neighbors. Thus G is strongly regular $\mathrm{SRG}(n, d, \lambda, \mu)$, and d equals the positive root $\frac{n}{4}\big(a + \sqrt{a^2 + 4\mu(n-1)}\big)$ of the quadratic (equivalently, the displayed extremal bound is tight for that parameter set). Such parameter tuples need not be realizable; equality occurs only for graphs in this extremal class when they exist. e.g. $C_5 = (5, 2, 0, 1)$. □

Exercise 7.3.6. *Let G be a graph with n points and m edges that does not contain a $K_{r,r}$. Prove that*

$$m < C \cdot n^{2-\frac{1}{r}},$$

where C only depends on r (not n or m). (Hint: Count $K_{1,r}$'s in two ways.)

证明. Let $\mathcal{M}$ be the set of pairs (v, S) where $v \in V(G)$, $S \subseteq N_G(v)$, and $|S| = r$. Then $|\mathcal{M}| = \sum_{v\in V} \binom{d(v)}{r}$.

Fix an r-element set $U \subseteq V$. If $U \subseteq N_G(v)$, then v is adjacent to every vertex of U, i.e. $v \in \bigcap_{u\in U} N_G(u)$. Since G has no $K_{r,r}$, the set U has at most $r-1$ common neighbors, so at most $r-1$ vertices v can occur with this U. There are $\binom{n}{r}$ choices of U, hence

$$\sum_{v\in V} \binom{d(v)}{r} \le (r-1)\binom{n}{r}. \tag{7.2}$$

For real $t \ge r-1$, set $\binom{t}{r} := t(t-1)\cdots(t-r+1)/r!$. The function $t \mapsto \binom{t}{r}$ is convex on $[r-1, \infty)$ (its second derivative is nonnegative). Thus, among integer degree sequences $(d(v))_{v\in V}$ with fixed sum $\sum_v d(v) = 2m$, the quantity $\sum_v \binom{d(v)}{r}$ is minimized when all degrees are as equal as possible, so

$$\sum_{v\in V} \binom{d(v)}{r} \ge n\binom{2m/n}{r}.$$

Combining with (7.2) gives

$$n\binom{2m/n}{r} \le (r-1)\binom{n}{r}. \tag{7.3}$$

If $2m \le n(r-1)$, then $m \le \frac{r-1}{2}n$ and the desired bound holds once C is large enough (depending on r only). Assume $2m > n(r-1)$. Then each factor in $\binom{2m/n}{r}$ is positive and at least $2m/n - (r-1)$, hence

$$\binom{2m/n}{r} \ge \frac{1}{r!}\Big(\frac{2m}{n} - (r-1)\Big)^r.$$

Also $\binom{n}{r} \le n^r/r!$. Substituting into (7.3) yields

$$n\Big(\frac{2m}{n} - (r-1)\Big)^r \le (r-1)n^r, \qquad \text{i.e.} \qquad \Big(2m - n(r-1)\Big)^r \le (r-1)n^{2r-1}.$$

Taking rth roots and rearranging,

$$m \le \frac{n(r-1)}{2} + \frac{(r-1)^{1/r}}{2}\, n^{2-1/r}.$$

Therefore $m \le \big(\frac{r-1}{2} + \frac{(r-1)^{1/r}}{2}\big)n^{2-1/r}$ once n is large enough that the right-hand side dominates the case $2m \le n(r-1)$. Choosing any fixed

$$C > \frac{r-1}{2} + \frac{(r-1)^{1/r}}{2}$$

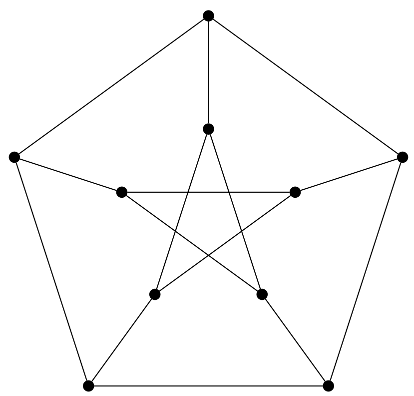
and then increasing C finitely many times if needed to cover finitely many small n, we obtain $m < C\, n^{2-1/r}$ for all n. $\square$

第八章　Quiz

8.1　Quiz 1

Exercise 8.1.1. *Give a graph with degree* 3*, girth* 5*, and* 10 *vertices.*

证明. The Petersen graph (below) has 10 vertices, is 3-regular, and has girth 5.



□

Exercise 8.1.2. *Suppose that G is a regular graph on* 10 *vertices with adjacency matrix A,*

$$\operatorname{tr}(A^2) = 30, \qquad \operatorname{tr}(A^3) = 0, \qquad \textit{and} \qquad \operatorname{tr}(A^4) = 150.$$

(a) How many triangles does G have?

(b) How many 4*-cycles does G have?*

证明. Let $n = 10$ and let G be k-regular.

(a) For any graph, $(A^2)_{ii} = \deg(i)$. Hence

$$\operatorname{tr}(A^2) = \sum_{i=1}^{10} (A^2)_{ii} = \sum_{i=1}^{10} \deg(i) = 10k = 30,$$

so $k = 3$.

Next, $(A^3)_{ii}$ counts closed walks of length 3 starting and ending at i. If $\{i, j, \ell\}$ spans a triangle, then exactly two such walks at i are $i - j - \ell - i$ and $i - \ell - j - i$, and there are no other closed 3-walks at i in a simple graph. Each triangle is counted at three vertices, each contributing 2 walks, so with T triangles, $\operatorname{tr}(A^3) = 6T$. Thus $T = \operatorname{tr}(A^3)/6 = 0$.

(b) We count closed walks of length 4 from a fixed vertex i and sum over i. Write a walk as $v_0 - v_1 - v_2 - v_3 - v_4$ with $v_0 = v_4 = i$.

Type I: $v_1 = v_3$. Then the walk is $i - a - v_2 - a - i$ with $a \in N(i)$. If $v_2 = i$ this is $i–a–i–a–i$, and there are $\deg(i) = 3$ choices for a. If $v_2 \neq i$ then $v_2 \in N(a) \setminus \{i\}$; since G is triangle-free, a has two neighbors other than i, so for each of the 3 choices of a there are 2 choices of v_2. Hence Type I contributes $3 + 6 = 9$ walks from i.

Type II: $v_2 = i$ and $v_1 \neq v_3$. Then the walk is $i–v_1–i–v_3–i$ with distinct $v_1, v_3 \in N(i)$, giving $3 \cdot 2 = 6$ walks.

Types I and II are disjoint as sequences of vertices. Any closed 4-walk using four *distinct* vertices $i - x - y - z - i$ lies on a (simple) 4-cycle through i and contributes at least 2 additional walks from i (the two directions around that cycle).

Therefore $(A^4)_{ii} \geq 9+6 = 15$, with equality if and only if i lies on no 4-cycle. Summing,

$$\operatorname{tr}(A^4) = \sum_{i=1}^{10} (A^4)_{ii} \geq 10 \cdot 15 = 150,$$

and equality holds only if every vertex lies on no 4-cycle, i.e. G contains no subgraph isomorphic to C_4. Since $\operatorname{tr}(A^4) = 150$, the number of (distinct) 4-cycles is 0. $\square$